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ABSTRACT

Title of Dissertation: The Effects of Organizational Characteristics on Entry
Timing: A Multi-Industry Study

Cher-Min Fong, Doctor of Philosophy, 1995

Dissertation directed by: Kenneth G. Smith, Professor and Stefan Wally, Assistant
Professor, College of Business and Management

This dissertation examined the relationships between organizational characteristics and firms' timing of entry into emerging product markets. This study is an extension of a stream of research on endogenous entry models. Research on this area essentially holds that not all firms have the same entry skills and costs. Therefore, a firm's entry timing decision is attributable to factors internal to the organization. In this dissertation, specific attention is directed toward the effects of top management team (TMT) demography, corporate governance, level of diversification, boundary spanning, past performance, capital structure, and firm size on entry timing. These organizational characteristics have been shown to be critical determinants of timing in the strategic management literature, but, remain almost unexplored in the entry timing research. Thus, through including these organizational characteristics, this study extends the current knowledge which is critical for answering the question: Why do some firms enter emerging product markets earlier/later than others?

The hypotheses relating organizational characteristics to entry timing are tested on data on ten product categories with 118 entrants. Top management team demographic data were gathered from Standard & Poor's Register of Corporations.

Directors and Executives and Dun & Bradstreet's Reference Book of Corporate Managements. The data for other organizational characteristics, including corporate governance, level of diversification, boundary spanning, firm performance, capital structure, and firm size were gathered from Moody's Industrial Manual and Public Utility Manual, Standard & Poor's Register of Corporations, Directors and Executives, and Dun's Million Dollar Directory.

To formulate a model of factors that influence the number of years it takes a contemplating firm to enter an emerging product market, this dissertation used the statistical framework of accelerated event-time analysis, an event-history approach. The advantage of event-history analysis is that it takes account of both the occurrence and timing of an event while estimating the effects of exogenous factors. Analyses were performed using the LIFEREG procedure in SAS.

The results of accelerated event-time analysis indicate that the firms that have less educated and more homogeneous TMTs, lower performance, greater boundary spanning, lower debt ratio, and are of larger size enter emerging product markets earlier. The practical implications of the results are discussed.

**THE EFFECTS OF ORGANIZATIONAL CHARACTERISTICS ON
ENTRY TIMING: A MULTI-INDUSTRY STUDY**

by

Cher-Min Fong

**Dissertation submitted to the Faculty of the Graduate School
of The University of Maryland in partial fulfillment
of the requirements for the degree of
Doctor of Philosophy
1995**

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CHAPTER 1

INTRODUCTION

As technologies change and market environments evolve, new products are introduced to create or satisfy new market demands or to substitute for existing products. A key decision for firms contemplating entry into emerging product markets is the timing of their entry. Firms may choose to enter earlier and hope to preempt critical strategic resources, but, they take the risk of premature entry. On the other hand, firms may prefer to enter later and wait for technical and market uncertainties to subside, but, they face the problem of being frozen out of a profitable market by participants who took the risk of early entry (Lieberman & Montgomery, 1988; Mitchell, 1989, 1991; Lilien & Yoon, 1990; Robinson, Fornell & Sullivan, 1992). Because firm's timing of entry into an emerging product market is such a dilemma, and of critical importance for the new product success or failure, it has received much attention in the literature.

Although there have been persistent research efforts aimed at providing a better understanding of the factors affecting entry timing, research in this area remains incomplete. It is not known, for instance, how a firm's top management team affects its timing of entry into an emerging market. Nor have the effects of board of directors, level of diversification, magnitude of financial resources, past performance, and firm size been investigated. This dissertation is directed to this

gap. To answer the question, What are the factors and how do they affect entry timing, this research develops a model linking managerial characteristics, corporate governance, operational domain, capital structure, past performance, and firm size to entry timing, and describes statistically the linkage based on a sample of ten industries with one hundred and eighteen entrants.

How Is Entry Decision Made: Review of Two Streams of Research Regarding Entry Timing

Two research streams stand out prominently in the studies on entry timing. One theme has centered on the risks and benefits associated with a firm's timing of entry into an emerging market. A variety of studies have been conducted to explain how advantage/disadvantage associated with particular entry timing decisions (e.g., first-mover or late entrant) and the sources of the advantage/disadvantage (e.g., technological leadership or leapfrogging) (Lieberman & Montgomery, 1988; Mitchell, 1991; Robinson, 1988; Robinson & Fornell, 1985; Lilien & Yoon, 1990). The other issue of interest is the rationale for entry at various times and under a variety of conditions. Such questions ask why some firms enter markets earlier than others, or what environmental, structural or managerial factors are associated with various entry decisions (Mitchell, 1989; Robinson, Fornell & Sullivan, 1992; Wernerfelt & Karnani, 1987).

Previous work on the first stream of research question has been based on the general argument that the choice of market-entry time is one of the major reasons for new product success or failure. Researchers of this area argue that the risks and opportunities of a new product vary due to changes in the general economy, changes in customer preferences, and evolution of the product life cycle (Lilien& Yoon, 1990; Lieberman & Montgomery, 1988). Thus, the decision to enter an emerging product market should be timed to balance the risks of premature entry and the possible missed opportunities due to entry too late. For example, in Robinson and Fornell's (1985) study, significant degrees of lateness effects on performance are found. They classify entrants into first-mover, pioneer, early follower and late follower. Their findings show that first-movers have higher market shares than later entrants. Early followers are also found to have higher market shares than late followers; however, the difference is much smaller than the difference between first-movers and early followers. The research of Bond and Lean (1977) gives another example for this research stream. Their findings show that the first-movers enjoy a substantial and enduring sales advantage. However, later entrants can overtake the pioneers by offering consumers significant new benefits. This research stream can be further exemplified by Srinivasan's (1988) study of pioneering versus early following. Generally, early followers are found to have marginally lower product quality and market shares than pioneers; however, early followers in product-markets in the initial stage of their life cycle have higher product quality and larger market shares. Further, early followers in the initial

phase of a product-market life cycle are more profitable than pioneers because of higher market shares and lower marketing and R&D expenses.

In the above research, entry timing is treated as exogenous ; that is, entry timing is hypothesized as a predictor variable determining performance (Moore, Boulding & Goodstein, 1991). Based on such models, firms are expected to enter emerging product markets at the timing that can maximize their performance. Among 13 empirical studies predicting the effects of entry timing/order on performance reviewed by Kerin, Varadarajan and Peterson (1992), a great majority of the prior research found the phenomenon of first-mover advantage. Generally, first-movers, although moderated somewhat by other factors, such as industry type, market conditions, or governmental regulation, perform better in terms of market shares, product quality, and cost position than later entrants. While this research stream provides us with insights regarding entry timing and subsequent risks and benefits, it raises the question of why, if pioneering improves firm performance, all firms do not attempt a first-entry strategy. The exogeneity research stream provides no information on answering this question.

To deal with the question of why all firms do not pursue the same entry timing strategy which is shown to be able to maximize their performance in the exogenous entry timing models, the second stream of entry timing research emphasizes that firms are essentially different in entry costs and skills. In

Lieberman and Montgomery's (1988) review article, the first issue they discuss is the endogeneity of entry decision -- that is, the notion that entry timing is a firm choice. In the Lieberman and Montgomery framework, firms assess their internal skills, scan the environment, and form expectations about future performance and the way in which it depends on entry timing in formulating their entry strategy.

Moore, Boulding and Goodstein (1991) develop a theoretical model of entry decision that suggests treatment of entry timing as an endogenous variable. They argue that if entry timing is endogenous, the decision is made fundamentally attributable to firm's proficiency and luck, rather than considering solely advantage/disadvantage which, in exogenous entry models, is entirely determined by entry timing, product life cycle and general economic factors. To provide empirical evidence for their model, Moore, Boulding and Goodstein reestimate Robinson and Fornell's (1985) model by relaxing the exogeneity assumption. The results of the reestimation confirm that entry timing should be treated as endogenous. They conclude that entry timing reflects a mixture of the effects of costs and skills difference and the effect of luck (Moore, Boulding & Goodstein, 1991).

The second stream of research question departs from the first research stream by more strongly focusing on the endogeneity issue. Based on the endogeneity assumption, research on this stream essentially holds that not all firms have the same entry costs and skills; therefore, a firm's entry timing decision is

attributable to factors internal to the organization. Researchers have linked a firm's specific assets and functional capabilities to entry timing (Teece, 1986; Wernerfelt & Karnani, 1987; Mitchell, 1989; Robinson, Fornell & Sullivan, 1992). Teece (1986) argues that due to the possession of complementary assets, a fast second or even a slow third may outperform the first entrant. He points out that in addition to core technological know-how in product innovation, complementary assets, such as marketing services, competitive manufacturing and after-sale support, are critical for new product success or failure. Larger firms often have less significant records with respect to innovation; however, they are more likely to possess the relevant assets. Because of this advantage, they may choose to enter later and still can easily outdo smaller earlier entrants. Wernerfelt and Karnani (1987) make a similar argument that a firm which commands relevant resources, such as financial resources, strong reputation and good cost position, all in sufficiently large quantities, may be able to enter late and yet snatch the leadership position from weaker early entrants.

Mitchell (1989) argues that a firm's entry timing is influenced by threats of the emerging product to the firm's core products and the possession of industry-specialized assets. The greater the threats of the emerging product, the earlier the firm will enter the market. Further, he argues that if there are only a few incumbents dominating an industry's specialized assets, these firms tend to enter an emerging subfield slowly, in order to retain the value of their assets. However,

when many firms possess industry-specialized assets, the firm possessing specialized assets will enter the emerging subfield earlier to occupy a better position. His study of the diagnostic imaging industry supports these arguments.

Drawing largely from the research of Lieberman and Montgomery (1988), Porter (1980), Lambkin (1988) and Lilien and Yoon (1990), Robinson, Fornell and Sullivan (1992) predict a relationship between functional capabilities and entry timing, which is categorized into first-mover, pioneer, early follower and late entrant. They argue that R&D skill is the key function in the early stage of the product life cycle, manufacturing ability is critical for early followers, and marketing capability is important for late entrants who enter in the maturity stage. Robinson, Fornell and Sullivan's empirical results show no significant difference between early and late entrants in R&D skill. However, late entrants are found to be stronger in marketing and manufacturing skills than early entrants (pioneer and first-mover).

The above work of the second research stream has provided a useful start for understanding the endogeneity of entry timing ; that is, entry timing as a firm's strategic decision. However, the relationships between entry timing and many organizational characteristics, which have been shown to be critical determinants for strategic decision making, remain unexplored. The organizational characteristics permeating the strategic management literature may include top

management team demography (Hambrick & Mason, 1984; O'Reilly & Flatt, 1989; Bantel & Jackson, 1989; Grimm & Smith, 1991; Wiersema & Bantel, 1992; Michel & Hambrick, 1992), corporate governance (Hill & Snell, 1988; Baysinger & Hoskisson, 1990; Hoskisson & Hitt, 1990; Johnson, Hoskisson & Hitt, 1993; Beatty & Zajac, 1994; Smart & Hitt, 1994; Hoskisson, Johnson & Moesel, 1994; Daily & Dalton, 1994), level of diversification (Baysinger & Hoskisson, 1989; Hitt, Hoskisson & Ireland, 1990; Hoskisson, Hitt & Hill, 1991; Hoskisson & Johnson, 1992; Hoskisson, Johnson & Moesel, 1994), organizational performance (Cyert & March, 1963; Manns & March, 1978; Kahneman & Tversky, 1979, 1981; Bourgeois, 1981; Singh, 1986; Chakravarthy, 1986; March & Shapira, 1987), boundary spanning capability (Dollinger, 1984; Gannon, Smith & Grimm, 1991; Smith, Grimm & Gannon, 1992; Sutcliffe, 1994), capital structure (Williamson, 1988; Baysinger & Hoskisson, 1989; Hitt, Hoskisson & Ireland, 1990; Long & Ravenscraft, 1993; Board, Delargy & Tonks, 1993) and firm size (Scherer & Ross, 1990; Hitt, Hoskisson & Ireland, 1990; Fiegenbaum & Karnani, 1991). Table 1 shows how previous research has linked these organizational characteristics to various strategic management issues.

**Table 1 Summary of Organizational Characteristics --
Strategic Management Research**

Organizational
Characteristics

Strategic Management Issues

Top Management Team Demographic Characteristics	<u>Restructuring</u> (Smart & Hitt, 1994). <u>Strategic Persistence, Change and reorientation</u> (Finkelstein & Hambrick, 1990; Grimm & Smith, 1991; Lant, Milliken & Batra, 1992; Wiersema & Bantel, 1992). <u>Diversification</u> (Michel & Hambrick, 1992; Hambrick & Mason, 1984). <u>Innovation and Emphasis on R&D</u> (Kimberly & Evanisko, 1981; Hambrick & Mason, 1984; O'Reilly & Flatt, 1989; Bantel & Jackson, 1989; Smith, Smith, Sims, Olian & Scully, 1994). <u>Risk Taking</u> (Gupta & Govindarajan, 1984). <u>Competitive Action and Response</u> (Smith, Grimm & Gannon, 1992).
Corporate Governance	<u>Restructuring</u> (Hoskisson & Turk, 1990; Johnson, Hoskisson & Hitt, 1993; Hoskisson, Johnson & Moesel, 1994; Smart & Hitt, 1994). <u>Strategic Persistence, Change and Reorientation</u> (Goodstein, Gautam & Boeker, 1994). <u>Diversification</u> (Hill & Snell, 1988; Baysinger & Hoskisson, 1990). <u>Innovation and Emphasis on R&D</u> (Hill & Snell & Snell, 1988; Baysinger & Hoskisson, 1990). <u>Risk Taking</u> (Beatty & Zajac, 1994).
Level of Diversification	<u>Restructuring</u> (Hoskisson & Turk, 1990; Hoskisson, Hitt & Hill, 1991). <u>Innovation and Emphasis on R&D</u> (Hoskisson & Hitt, 1988, 1990; Baysinger & Hoskisson, 1989; Hitt, Hoskisson & Ireland, 1990; Hoskisson & Johnson, 1992). <u>Risk Taking</u> (Hoskisson & Hitt, 1990; Hoskisson, Hitt & Hill, 1991).
Organizational Performance	<u>Strategic Persistence, Change and Reorientation</u> (Finkelstein & Hambrick, 1990; Hambrick, Geletkanycz & Fredrickson, 1993). <u>Innovation and Emphasis on R&D</u> (Bourgeois,

	1981; Manns & March, 1978). <u>Risk Taking</u> (Kahenman & Tversky, 1979; Singh, 1986; March & Shapira, 1987). <u>Strategic Decision Making</u> (Cyert & March, 1963; Cohen, March & Olsen, 1972).
Boundary Spanning Activities	<u>Competitive Action and Response</u> (Smith, Grimm & Gannon, 1992). <u>Innovation and Emphasis on R&D</u> (Quinn, 1985; Kanter, 1988). <u>Adaptation to Environment</u> (Dollinger, 1984; Sutcliffe, 1994).
Capital Structure	<u>Restructuring</u> (Long & Ravenscraft, 1993). <u>Diversification</u> (Hitt, Hoskisson & Ireland, 1990). <u>Innovation and Emphasis on R&D</u> (Baysinger & Hoskisson, 1989; Hitt, Hoskisson & Ireland, 1990; Board, Delargy & Tonks, 1993; Long & Ravenscraft, 1993).
Firm Size	<u>Innovation and Emphasis on R&D</u> (Roberts, 1965, 1988; Teece, 1986; Baysinger & Hoskisson, 1989; Hitt, Hoskisson & Ireland, 1990; Scherer & Ross, 1990). <u>Output Flexibility</u> (Fiegenbaum & Karnani, 1991).

Overview of This Dissertation

Although the organizational characteristics depicted in Table 1 have been well documented in the strategic management literature, their effects on entry timing remain poorly understood. The absence of these factors in entry timing

research may essentially hinder our knowledge of a firm's entry behavior. Thus, the purpose of this dissertation is to extend the endogeneity research stream through the development and empirical test of a model linking entry timing and these organizational characteristics. The organizational factors examined herein focus on: top management team, corporate governance, levels of diversification, firm performance, capital structure, boundary spanning capacity, and firm size. As discussed above, through including these unexplored organizational characteristics, this dissertation extends the current knowledge which is critical for answering the following question:

Why do some firms enter emerging markets earlier than others, or what endogenous organizational factors are associated with entry timing?

In sum, this dissertation investigates the relationships between a firm's timing of entry into an emerging product market and organizational characteristics, which have been widely documented to be important determinants of strategic management issues, however, have not been explored vis-a-vis entry timing. This research direction is of theoretical interest, and this study responds to the suggestions for such research by previous investigators (see Lieberman & Montgomery, 1988; Moore, Boulding & Goodstein, 1991). Furthermore, this dissertation has important implications for practitioners. It gives managers new insights for analyzing their own organization's characteristics and intended strategic

decisions relating to entry timing. It also provides a base for analyzing competitors' entry timing.

Organization of the Remainder of the Dissertation

This dissertation has six chapters. Chapter 2 introduces two strategic tendencies: a firm's degree of focusing on innovation and its attitude toward risk taking. These two tendencies are used to link organizational characteristics and entry timing. The strength of these two strategic tendencies will influence firm's entry timing decisions. The research model for this dissertation is then introduced.

Chapter 3 builds the theoretical framework for the research and develops hypotheses for empirical testing. Key concepts and terminology used are explained and defined. The organizational characteristics examined in this research include: top management team demographic traits and diversity, corporate governance, organizational slack, boundary spanning activities, capital structure, and firm size. For each of these organizational characteristics, a theoretical argument is made concerning the relationships of organizational characteristics --> strategic tendencies --> entry timing. This is followed with the hypothesis presenting the expected relationships for empirical testing. For three organizational

characteristics, top management team diversity, organizational slack and firm size, there appear to be alternative lines of argument leading to conflicting predictions. Competing hypotheses are developed for these organizational characteristics. Chapter 3 ends with a discussion of four control variables: incumbent vs. newcomer, potential competitive threat, market size incentive, and industry specific entry behavior.

Chapter 4 presents the research methodology for the dissertation. The criteria for the selection of products are discussed. The data source and data collection methods are presented. This is followed with operational definitions and units of measure for each of the dependent and independent variables. Accelerated event-time analysis is used to test the hypotheses. The general method and baseline distributions of accelerated event-time analysis are discussed. This is followed by the discussion of life-table analysis which is used to estimate the survival functions.

Chapter 5 presents the research results. Descriptive statistics and correlation matrix of variables are shown. The estimated survival functions and the estimation process are discussed. Finally, the results of accelerated event-time analysis are shown for the tests of the hypotheses predicting the relationships between organizational characteristics and entry timing.

The final chapter discusses the research results. Specifically, the effects of examined organizational characteristics on entry timing are explained.

Recommendations are made for continuing research. This research ends with its managerial implications and the dissertation conclusions.

CHAPTER 2

INNOVATION AND RISK-TAKING: TWO STRATEGIC DIMENSIONS LINKING ORGANIZATIONAL CHARACTERISTICS TO ENTRY TIMING

Miles and Snow (1978) suggest that strategy is a pattern or stream of major and minor decisions about a firm's possible future domains. In this research the term, strategic dimension is used to represent the pattern or stream of firm decisions. For examining organizational characteristics - entry timing linkage, this research employs two strategic dimensions: (1) a firm's degree of focusing on innovation, and (2) a firm's attitude toward risk. These two strategic dimensions will not be specifically measured and tested in this dissertation, but, with substantial theoretical foundations, they will play important roles explaining why organizational characteristics are linked to entry timing. As will be discussed below, decisions relating to entry timing are influenced by the strength of these two dimensions.

Degree of Focusing on Innovation

Earlier entrants are more likely to confront conditions of less paradigmatic product design and a fast changing market environment (Teece, 1986; Schnaars, 1986). Extending Kuhn's (1970) concept of scientific evolution to the notion of

product development, Teece (1986) argues that there are two stages in the technological evolution of an industry: (1) the pre-paradigmatic stage, when no generally accepted product design emerges, and (2) the paradigmatic stage, which begins when a generally accepted product design emerges. The product design that eventually will dominate the market is unclear during the early stages of an emerging product market (Porter, 1980; Teece, 1986; Schnaars, 1986). For gaining the dominant position or avoiding being leapfrogged by later entrants with better product design, Teece (1986) argues that product innovation is the most critical competitive measure during the pre-paradigmatic stage. However, "at some point in time and after considerable trial and error in the marketplace, one design or a narrow class of design begins to emerge the more promising" (p.190). Later entrants can imitate the fundamental design and make only incremental changes. Competitive focus shifts as an industry evolves (Porter, 1980; Teece, 1986). Product innovation may gradually lose its competitive criticality when the industry becomes more mature. Thus, a firm's degree of focusing on innovation is the first indicator of entry timing. The firms that have higher degrees of emphasis on innovation enter emerging product markets earlier.

Attitude Toward Risk-Taking

The risks earlier entrants may confront include free-rider effects, market

and technological uncertainty and shifts in technology or customer needs (Porter, 1985; Lieberman & Montgomery, 1988). Later entrants may be able to free-ride on pioneering firms' investments in R&D, buyer education, and infrastructure development at lower entry costs. Earlier entrants may be vulnerable if buyer needs change and their technologies are no longer valued. Earlier entrants may be inflexible in responding to major shifts in technology and market demand because of being locked in a specific set of old investments. Furthermore, many growth opportunities may turn out to be illusory (Schnaars, 1986). When firms finally find out that they have entered a market of poor prospect, these chimeras may have become "traps for cash, management attention, and other resources that , in hindsight, could have been deployed in more profitable endeavors" (Schnaars, 1986: p.32). These uncertainties constitute a highly risky environment for earlier entrants. Therefore, a firm's attitude toward risk-taking forms the second indicator of entry timing. Risk takers will be earlier entrants.

Other strategic dimensions could also be posited, but these two, as discussed, have substantial theoretical foundations in previous entry timing research and act as the theoretical connection explaining how organizational characteristics are linked to entry timing. As depicted in Figure 1, the examined organizational characteristics are important determinants of these two strategic dimensions. The effects of these organizational characteristics on innovation and risk taking and, ultimately, on entry timing will be elaborated in the following chapter.

CHAPTER 3

THEORY AND HYPOTHESES

The focus of this dissertation is on the effects of organizational characteristics on entry timing. Figure 1 shows the overview of the research model for this dissertation. In this chapter, the rationale behind the hypotheses is explained and the hypotheses are presented for testing.

Traits of Top Management Team

The perspective developed by Hambrick and Mason (1984) and others (e.g., O'Reilly & Flatt, 1989; Bantel & Jackson, 1989; Grimm & Smith, 1991; Smith, Grimm & Gannon, 1992; Wiersema & Bantel, 1992; Smith, Smith, Olian, Sims, O'Bannon & Scully, 1994; Smith, Smith, Sims, Olian & Scully, 1994) suggests that the demographic characteristics of the top management team (TMT) affect strategic decisions. Thus, this perspective implies that TMT traits may affect a firm's degree of focusing on innovation and attitude toward risk, and thereby affect entry timing. In this section, three TMT traits, age, education, and organizational tenure, are examined to uncover the relationship.

TMT Age

The age of the TMT is expected to influence a firm's degree of focusing on innovation and attitude toward risk (Hambrick & Mason, 1984; Grimm & Smith, 1991; Smith, Grimm & Gannon, 1992; Wiersema & Bantel, 1992; Smith, Smith, Sims, Olian, Scully, 1994). Research suggests that flexibility decreases as people age, while rigidity and resistance to accept new things increase. Hambrick and Mason (1984) argue that because younger managers are more likely to seek growth, they tend to pursue more innovative and risky strategies. On the other hand, older managers are more conservative than their younger counterparts. Older managers are less innovative because they are less able to grasp new ideas and learn new behaviors (Hambrick & Mason, 1984; Bantel & Jackson, 1989). Further, older managers tend to avoid risky decisions because, to them, security, both financial and career-related, becomes much more important (Wiersema & Bantel, 1992). Older managers presumably have accumulated more prestige and wealth relative to younger managers and, consequently, are expected to engage in less risky behavior to avoid reducing their status or wealth (Smart & Hitt, 1994). Grimm and Smith (1991), in their railroad study, found that older managers would be more committed to the status quo, which means a lessening of aggressive strategic behavior. Also, older managers are found to be negatively related to innovation in Bantel and Jackson's (1989) banking study.

TMT Education

According to Hambrick and Mason (1984), education may be an indicator of a manager's knowledge and skill base. Typically, because product innovation is interdisciplinary, it benefits from broader perspectives and information from outside (Kanter, 1988; Quinn, 1985). Managers with higher levels of education are associated with a higher capability for information processing and the ability to discriminate among a variety of stimuli (Wiersema & Bantel, 1992). Educated individuals are likely to engage in more boundary spanning activities as well as to show a greater willingness to interact with the environment (Wiersema & Bantel, 1992; Gannon, Smith & Grimm, 1991). Therefore, because of their information processing capability and outward orientation, managers with higher levels of education are expected to head more innovative organizations. Further, when facing a risky environment, more educated managers are more likely to have a higher tolerance for ambiguity and show a higher ability for integrative complexity (Wiersema & Bantel, 1992; Dollinger, 1984). Thus it seems reasonable to expect that greater levels of education would be associated with risk taking.

TMT Organizational Tenure

Researchers have shown that if top managers have spent their careers in one industry and one organization, they will have a limited knowledge base from which

to conduct a strategic search for new opportunities and would not be likely to pursue new ideas outside this knowledge base (Hambrick & Mason, 1984; Grimm & Smith, 1991; Smith, Grimm & Gannon, 1992). Smart and Hitt (1994) argue that managers gain firm-specific knowledge and experience as their organizational tenure increases. As firm specific knowledge increases, the employment mobility of managers decreases. Because they have fewer employment opportunities, managers with longer organizational tenure may be more conservative than managers with less organizational tenure. Furthermore, organizational tenure is associated with a higher commitment to the status quo in an organization which implies that long tenured managers are less likely to employ strategies involving new terrain, e.g., product innovation (Hambrick & Mason, 1984; Smart & Hitt, 1994; Smith, Grimm & Gannon, 1992). Smith, Grimm and Gannon observed that less experienced managers are more likely to engage in risky and novel actions. They argue that it is because managers with less organizational experience are more likely to challenge standardized procedures and rules. Thus, it is expected that longer organizational tenure will be associated with less innovation and risk aversion.

In light of these arguments, the following hypotheses are proposed for the relationships between TMT traits and entry timing.

Hypothesis 1: The younger the age of the top management team, the earlier the firm will enter an emerging product market.

Hypothesis 2: The higher the educational levels of the top management team, the earlier the firm will enter an emerging product market.

Hypothesis 3: The fewer the years of inside service of the top management team, the earlier the firm will enter an emerging product market.

Diversity of Top Management Team

Previous research has widely documented the relationship of TMT demographic homogeneity/heterogeneity and organizational innovation. Some researchers simultaneously include two competing predictions relating TMT demographic diversity to organizational innovation in their studies (O'Reilly & Flatt, 1989; Bantel & Jackson, 1989). On the one hand, because innovation requires effective integration of various functions, TMT homogeneity is expected to be positively associated with organizational innovation. On the other hand, innovation demands broader perspectives; therefore, TMT heterogeneity should be related to organizational innovation. These two opposing predictions relating TMT demographic diversity to organizational innovation and, subsequently, to entry timing, will be discussed in more detail below.

TMT Homogeneity

Because product innovation involves interdisciplinary or interfunctional effort, integration of functions is an important antecedent to this complex process (Kanter, 1985, 1988; Smith, Smith, Sims, Olian & Scully, 1994). Souder (1978), for instance, shows that when marketing and R&D are not effectively integrated at top level of management, innovation suffers. This integration is particularly important, since more than 70% of new product innovation ideas come from market factors and are realized by integrated works (Utterback, 1974). Therefore, specialization in separate staff functions may inhibit the communication needed for product innovation unless top management provides effective integrating mechanisms (O'Reilly & Flatt, 1989). The integration required to link together pieces of organization demands good interaction between individuals. A homogeneous TMT may enhance effective interaction and communication. In general, demographic similarity is associated with a higher likelihood of similar value, experience and personal attraction (Tsui & O'Reilly, 1989). Shared cohort membership, experiencing the same events, such as birth or organizational entry within the same time interval, indicates similar exposure to social, environmental and organizational events (Wiersema & Bantel, 1992; Smith, Smith, Olian, Sims, O'Bannon & Scully, 1994). The similarity may then be associated with more frequent and accurate communication and high levels of group cohesion. Therefore, from this point of view, homogeneity of TMT is expected to facilitate

the firm's overall ability to create innovative products.

TMT Heterogeneity

Research suggests that creative ideas often spring up at the boundaries of specialties and disciplines, rather than squarely in the middle (Kanter, 1988). The advantage of cross-fertilization of ideas is more likely to be associated with diverse teams. A heterogeneous team is expected to gather information from a variety of sources and to have diverse interpretations and perspectives (Wiersema & Bantel, 1992). A key benefit of a heterogeneous team is that its members' different points of view can generate diversity, novelty, and comprehensiveness in the set of recommended solutions (Bantel & Jackson, 1989; Wiersema & Bantel, 1992). Thus, from this point of view, TMT heterogeneity can stimulate organizational innovation.

Because these alternative lines of argument clearly lead to conflicting predictions about the relationship between innovation and TMT diversity, two competing hypotheses are proposed.

Hypothesis 4a: The greater the top management team homogeneity, the earlier the firm will enter an emerging product market.

Hypothesis 4b: The greater the top management team heterogeneity, the earlier the firm will enter an emerging product market.

Corporate Governance

Agency theory assumes that managers and shareholders desire to maximize their own interests (Baysinger & Hoskisson, 1990; Smart & Hitt, 1994). Therefore, if managers do not hold significant ownership in the firm, the interests of managers and owners may diverge. This is referred to as an agency problem -- that is, managers may act in their own self-interests and produce costs to shareholders (Eisenhardt, 1989; Beatty & Zajac, 1990; Smart & Hitt, 1994). It is argued that agency costs can be constrained or attenuated by corporate governance (Fama & Jensen, 1983; Baysinger & Hoskisson, 1990). Corporate law grants boards of directors the formal authority, on the basis of reflecting shareholders' interests, to hire, fire, evaluate, and compensate top management. Agency theorists argue that the board with its legal authority, is an important element of corporate governance (Fama & Jensen, 1983; Williamson, 1984). In this section, specific attention is directed toward the potential effects of two board related governance issues, board composition and duality of CEO and chairperson positions, on firm's innovation and risk taking and, ultimately, entry timing.

Composition of Boards

The distinction between outside and inside directors has been judged to be critical in corporate governance (Baysinger & Hoskisson, 1990; Hoskisson & Turk, 1990; Johnson, Hoskisson & Hitt, 1993; Smart & Hitt, 1994; Westphal & Zajac, 1994). Research suggests that inside directors who hold managerial positions in the firm have relatively greater access to the daily decision process of the firm's top management (Baysinger & Hoskisson, 1990; Hoskisson & Turk, 1990). Thus, they are more likely to employ strategic controls, which are characterized as the control mechanisms that rely on subjective information for assessing the quality of decision-making processes (Gupta, 1987). In contrast, outside directors, due to limited contact with the day-to-day decision processes of the firm, usually enjoy few of these informational advantages. Therefore, outside directors are much less likely to use strategic controls (Baysinger & Hoskisson, 1990). They tend to employ financial controls under which managers are evaluated and rewarded primarily on the basis of their success in meeting objective financial performance criteria (Baysinger & Hoskisson, 1990; Hoskisson & Turk, 1990; Hoskisson, Johnson & Moesel, 1994).

Control theory suggests that the type of control strategy chosen affects both the intensity of managers' efforts and the direction of that effort (Hambrick & Snow, 1988). The firms with insider-controlled boards, being better able to

differentiate between decisions and risks, are more likely to evaluate and reward managers on the basis of strategic controls. Under such a system of controls, managers are more willing to take risks and focus on innovation, because they are less likely to have to be responsible for the financial adversity (ex post) which may be partly out of their control in the decision making process (ex ante). On the other hand, in firms where outsiders dominate the boards, managers are evaluated on the basis of financial criteria, with rewards contingent on these criteria. While individual investors can minimize their risks by manipulating investment portfolio, managers, in contrast, are not able to readily diversify their employment and financial risks (Baysinger & Hoskisson, 1990). Increasing managers' liability for financial performance in the short run generally not only increases their diligence in maximizing short-term profits at the sacrifice of long-term investments which are usually represented by R&D expenditures, but it also reduces their incentive to take risks (Fischel & Bradley, 1986).

Based on these arguments, the following hypothesis is proposed for the relationship between board composition and entry timing.

Hypothesis 5: The greater the ratio of inside directors to total board members, the earlier the firm will enter an emerging product market.

Duality of CEO and Board Chairperson Positions

Another board-related governance issue involves CEO duality. CEO duality occurs when the same person holds both the CEO and board chairperson positions in a corporation (Rechner & Dalton, 1991; Finkelstein & D'Aveni, 1994). As will be discussed in this section, a body of work in administrative theory and strategy formulation on the concepts of unity of command and centralization of power has important implications for CEO duality.

CEO duality, the consolidation of the two most senior management positions, establishes a unity of command at the top of firm, with unambiguous leadership and centralized power (Finkelstein & D'Aveni, 1994). CEO duality, by establishing clear lines of authority and responsibility within a firm, helps to avoid role and responsibility conflicts among managers. Research suggests that clear role and responsibility are critical for innovative firms (Schoonhoven & Jelinek, 1990). As Schoonhoven and Jelinek (1990:p.95) state, "(in these innovative, high technology firms) there is high predictability of who has responsibility for what throughout companies." Thus, based on this, CEO duality is expected to be positively related to firm's focusing on innovation.

Miller and Friesen (1977, 1984) argue that the diffusion of power makes it difficult for a firm to take any decisive actions. As discussed above, CEO duality

enhances the centralization of power. According to Miller (1987:p.13), centralization of power "can free top managers to be assertive -- venturesome and proactive -- because they have power to commit significant resources to a project." Further, the capability of quickly mobilizing resources and making decisions is critical for firms under risky situations. Research suggests that centralization of power can facilitate these quick actions by firms (Allison, 1971; Miller, 1987; Wally & Baum, 1994). Thus, it is reasonable to expect that CEO duality is positively related to a firm's attitude about risk taking.

From the proceeding arguments, the hypothesis for the relationship between CEO duality and entry timing is:

Hypothesis 6: A firm with duality of CEO and board chairperson positions will enter an emerging product market earlier.

Level of Diversification

The research linking firm's diversification to innovation and risk taking has been well documented (Hoskisson & Hitt, 1988; Baysinger & Hoskisson, 1989; Hoskisson & Turk, 1990; Hitt, Hoskisson & Ireland, 1990; Hoskisson, Hitt & Hill, 1991; Hoskisson & Johnson, 1992; Hoskisson, Johnson & Moesel, 1994). Hitt, Hoskisson and Ireland (1990) suggest that "this research can be understood

more clearly by examining how increases in diversification are managed through organizational controls" (p.34).

In less diversified firms, corporate executives often have thorough knowledge of their firms' operations. Thus, they are able to evaluate the plans and intended actions of business unit managers by using strategic controls (Baysinger & Hoskisson, 1989; Hitt, Hoskisson & Ireland, 1990; Hoskisson & Johnson, 1992). As discussed in the section on Corporate Governance, under a system of strategic controls, business unit managers are more likely to engage in innovative and risky projects because they know that the strategies they implement are understood by corporate executives and evaluated on their strategic applicability (Baysinger & Hoskisson, 1989; Hoskisson, Johnson & Moesel, 1994). However, when a firm becomes more diversified, both in terms of the number of divisions under corporate management control and in terms of relatedness among product divisions, it is increasingly difficult for corporate executives to process effectively the volume of information they receive. It is argued that as a multiproduct firm becomes more diversified, two conditions result (Hitt, Hoskisson & Ireland, 1990). First, the horizontal, managerial span of control increases. Second, the vertical separation from knowledge associated with operational decision-making increases. This may lead to limiting the corporate executives' abilities to gather, process and interpret the information needed to evaluate business unit performance accurately and allocate resources and rewards (Williamson, 1975; Hitt, Hoskisson & Ireland,

1990). Thus, they may tend to focus attention almost exclusively on financial outcomes -- especially the net profit for a period and the rate of return on invested capital (Ackerman, 1970; Bower, 1972; Hill & Hoskisson, 1987). Also, they tend to increasingly employ portfolio techniques, with allocation of resources based primarily on sources and uses of cash rather than strategic criteria (Hitt, Hoskisson & Ireland, 1990; Hoskisson, Hitt & Hill, 1991; Baysinger & Hoskisson, 1989). Under such system of financial controls, as discussed previously, business unit managers, due to increases in liability for short-term financial performance, not only are less likely to engage in long-term investments such as innovation projects, but they also tend to avoid risky strategies.

Based on these arguments, the hypothesis for the relationship between level of diversification and entry timing is therefore proposed.

Hypothesis 7: The lower the level of diversification, the earlier the firm will enter an emerging product market.

Organizational Performance

Two opposing lines of argument relating organizational performance to a firm's innovation and risk taking have been documented in the literature . One

school of thought argues that good performance generates organizational slack, which in turn can provide resources for innovation and risk taking (Rosner, 1968; Damanpour, 1987; Smith, Grimm & Gannon, 1992). The other school argues that when a firm is performing poorly, the desire to reach the performance target generally promotes greater risk taking and emphasis on innovation (Manns & March, 1978; Kahneman & Tversky, 1979; Singh, 1986; March & Shapira, 1987). The following section discusses these two contradictory points of view in more detail.

High Performing Firms

It is a generally acceptable concept that good performance is related to high organizational slack (Cyert & March, 1963; Manns & March, 1978; Singh, 1986; Chakravarthy, 1986). According to Cyert and March (1963), organizational slack grows in organizations that are performing well, but during adversity, slack reduces and aspiration levels are maintained. Therefore, slack plays an important role of absorbing fluctuations in organizational performance. Through influencing increases/decreases in organizational slack, firm performance is expected to have impact on innovation and risk taking.

Viewing slack as a resource in excess of what is required for the normal

operation of an organization, one can argue that its presence allows an organization to interact or compete in its environment more boldly (Bourgeois, 1981; Singh, 1986). As a firm performs well, slack is generated; the organization can afford to experiment with new strategies by entering new markets and/or introducing new products (Hambrick & Snow, 1977). Slack can be a buffer which helps a firm to maintain stability when facing adverse business situations (Cyert & March, 1963). Therefore, slack can encourage managers to take risks because it allows an organization the ability to absorb the costs associated with failure (Rosner, 1968; Singh, 1986). Organizational slack can also enhance innovation. Rosner (1968) notes that slack resources are able to help a firm to bear the costs of innovation and to explore new ideas in advance of actual need. Further, quality innovation benefits from ample information (Kanter, 1988; von Hippel, 1988). The high performing organizations with greater slack resources can afford sophisticated information search activities, such as an integrated computer information system to enhance search processes (Smith, Grimm & Gannon, 1992). Therefore, from this point of view, high performance can be expected to be related to innovation and risk taking.

In addition to generating greater organizational slack, high performance also creates "culture of pride" which stimulates risk taking and innovation (Kanter, 1988:p.182-183). Kanter argues that success breeds success; where there is a culture of pride, based on high past performance, people's feeling of confidence in

themselves and others goes up. They are more likely to take risks. Pride in high performing firms helps to facilitate innovation. A feeling that people inside the organization are competent leaders, that the organization has been successful because of its people, supports this.

Low Performing firms

Poor performance has been related to managerial risk taking and emphasis on innovation in a wide range of organizations (Manns & March, 1978; Kahneman & Tversky, 1979; Bowman, 1980, 1982; Singh, 1986; March & Shapira, 1987). March and Shapira (1987) argue that if performance is above a performance target, managers attempt to avoid actions that may produce below target performance; that is, for high performing managers, the dangers of falling below target performance dominate attention and the opportunities for gains are less important. Therefore, when a firm is performing well, decision makers are relatively risk averse. However, for decision makers who are below the performance targets, the strong intention to reach the targets focuses attention in a way that leads generally to greater risk taking (March & Shapira, 1987). This view is consistent with the predictions of prospect theory (Kahneman & Tversky, 1979; Tversky & Kahneman, 1981). Prospect theory predicts that the propensity of individuals to take risks is context dependent. While decision makers are risk-averse in the gains (above the reference level), they are often risk-seeking in the domain of losses

(below the reference level). Prospect theory has been widely supported by empirical evidence (Kahneman & Tversky, 1979; Singh, 1986; Fiegenbaum & Thomas, 1988). For example, Singh found that poor performance is related to high risk taking in organizational decisions and good performance is related to low risk taking.

Additionally, low performance has been related to greater organizational innovation. In their study of departments at Stanford University, Manns and March (1978) found that when the economic environment becomes more severe, departments need to increase income; one way of doing so is to introduce more innovations in their curricula. A department will be more likely to seek curriculum attractiveness and innovation in hard times; on the other hand, a department that can secure more funds is less dependent on maintaining innovative and attractive curriculum in order to compete for resources.

Furthermore, managers will not be receptive to change unless they are dissatisfied with the current situation (Kimberly & Quinn, 1984). When an organization is performing well, the managers tend to be committed to the status quo (Hambrick, Geletkanycz & Fredrickson, 1993), an attitude that is less likely to emphasize innovation. High performance levels create a confidence in the status quo; change is rarely considered. It is poor performance that erodes managerial confidence and leads to aggressive search for innovative methods and paradigms

(Cyert & March, 1963; Starbuck & Milliken, 1988; Hambrick, Geletkanycz & Fredrickson, 1993).

Based on the literature reviewed, two contrasting predictions relate organizational performance to innovation and risk taking. In light of this, competing hypotheses are therefore proposed.

Hypothesis 8a: The higher the organizational performance, the earlier the firm will enter an emerging product market.

Hypothesis 8b: The lower the organizational performance, the earlier the firm will enter an emerging product market.

Boundary Spanning

Product innovation processes, from idea generation to product design, are heavily dependent on external market and technical information (Kanter, 1988). Tushman (1977) contends that innovative organizations usually take better advantage of boundary spanners to bring them intelligence about their environments. Miles and Snow (1978) describe defenders as "organizations which are experts in their limited areas of operation but do not tend to search outside of their domains for new opportunities.....they allocate only a small amount of administrative time and personnel to monitoring other organizations, events and

trends" (1978:p.38). On the other hand, prospectors, the creators of change, have strong concern for product and market innovation. Miles and Snow (1978:p.56) state, "the prospector must develop and maintain the capability to monitor a wide range of environmental conditions, trends and events." Research on information-processing suggests that for managing environmental information, a firm must build an effective sensory system, or boundary spanning mechanism, which is actualized by units or individuals operating on the peripheries of the firm (Smith, Grimm & Gannon, 1992). Boundary spanners are critical to organizational innovation because they, standing on the border of the organization, categorize environmental information and package it in such a way that the organization can seize better opportunities for product innovation.

Boundary spanning activities are also expected to be related to risk taking. As discussed, the risk associated with early entry into an emerging market is partly a result of market and technological uncertainty. Duncan (1972) argues that managers feel uncertainty when there is a lack of information regarding environmental factors in the decision making situation. Based on this argument, given that every firm has identical tolerance for uncertainty and is exposed to the same objective environment, the firms with greater boundary spanning activities, because of their superior capability of environmental information processing, will perceive a less uncertain situation than that which their competitors may perceive. Therefore it is reasonable to expect that boundary spanning activities may facilitate

risk taking.

The following hypothesis states the expected relationship between boundary spanning activities and entry timing.

Hypothesis 9: The greater the boundary spanning activities, the earlier the firm will enter an emerging product market.

Capital Structure: Debt Position

Some previous studies have included debt position and the ability to raise debt in the discussion of a more generalized concept of slack resources (e.g. Bourgeois, 1981; Chakravarthy, 1986; Finkelstein & Hambrick, 1990). However, the central argument here is that different sources of capital will influence a firm's commitment to innovation and attitude toward risk differently. Ever since Modigliani and Miller (1958) argued that with perfect capital markets the source of financing is irrelevant, a large body of literature has arisen citing numerous reasons why this argument is incorrect, particularly for investment in R&D (e.g., Leland & Pyle, 1977; Smith & Warner, 1979; Williamson, 1988; Long & Ravenscraft, 1993). According to this literature, a firm's capital structure, especially its debt position, may have an impact on corporate behavior (Long &

Ravenscraft, 1993). Several theories which have been employed for predicting the relationship between debt position and firm's innovation and risk taking are discussed in the following section.

Numerous markets have been characterized by informational differences between buyers and sellers (Leland & Pyle, 1977). In financial markets, informational asymmetries between firm's management (sellers) and funding sources (buyers) are particularly pronounced. Insiders have superior information about their innovation projects, such as new product development, that is difficult to reveal to the capital market (Myers, 1984). Revealing this information may provide competitors a valuable signal and reduce the value of private information. Research suggests that it is difficult to disclose such information to external capital suppliers without competing firms becoming aware of this knowledge (Bhattacharya & Ritter, 1983). Thus for preventing the revelation of important information to competitors, internally generated funds are more appropriate for innovation projects than external funds (Long & Ravenscraft, 1993).

Debt financing of risky innovation investments is also difficult because of potential moral hazard problems (Leland & Pyle, 1977). In such cases, moral hazard refers to the agency problem that the borrower may exaggerate the value of their innovation project, but, the project is so complex that debt providers can not verify its true value (Eisenhardt, 1989). Because debt providers, unlike equity

holders, bear the downside risk and not the upside gains, they are likely to be hesitant to fund risky innovation projects (Leland & Pyle, 1977; Long & Ravenscraft, 1993). Thus as debtholder power increases, the firm's managers are less likely to undertake risky innovation projects.

Williamson (1988) proposes that debt operates largely through strict rules. He argues that these rules impose higher costs for projects where the assets involved are not easily redeployable for other purposes. Innovation regarding new product development generally involves assets that are highly specialized and firm specific (Long & Ravenscraft, 1993). Further, much of the capital involved in such projects is in the form of human capital, such as scientists and engineers, which can be characterized as non-redeployable assets. Thus, transaction cost theory suggests that debt financing is inappropriate for firms with innovation activities as key components of their competitiveness.

Organizational controls may also be used to explain the relationship between debt position and a firm's innovation and risk taking. As debt increases, corporate executives are forced to focus on short-term cash flow goals to meet interest payments (Long & Ravenscraft, 1993). Therefore, increased leverage is likely to lead to the use of financial controls. Under a system of financial controls, as discussed in the sections of Corporate Governance and Levels of Diversification, business unit managers are responsible for these cash flow requirements, and this,

in turn, produces managerial risk aversion and reduced commitment to innovation.

The following is a hypothesis for the expected relationship between capital structure and entry timing.

Hypothesis 10: The lower the level of debt in capital structure, the earlier the firm will enter an emerging product market.

Firm Size

The relationships between firm size and its innovation and risk taking have been well documented in the literature. However, this literature provides opposite predictions (Hitt, Hoskisson & Ireland, 1990; Scherer & Ross, 1990). One school of thought argues that economies of scale and scope allow large firms to be more efficient in development of innovation and more able to afford risky investments (Schumpeter, 1961). The other school of thought contends that large firms are more likely to use bureaucratic controls, which may frustrate valuable innovations even in the first place (Roberts, 1968). These two opposing points of view will be discussed in the next section.

Large Firms

Scherer and Ross (1990) argue that large firms have significant advantages in supporting research and innovation. Large size allows them to maintain a diversified portfolio of innovation efforts, hedging the risks that any given project will fail. The ability to exploit scale economies is another potential advantage for large firms (Schumpeter, 1961). Economies of scale in R&D activities allow large firms to be more efficient in the development of innovation, resulting in the generation of more innovative output at lower costs. Furthermore, large firms can employ specialists in many disciplines to cross-fertilize one another and offer temporary help when a team working on some project runs up against a technical problem outside its normal specialty (Scherer & Ross, 1990). Additionally, a large firm should have greater ability to commercialize innovative ideas (Schumpeter, 1961; Teece, 1986). That is, a large firm usually has well-established marketing channels, superior manufacturing systems, and better access to capital at lower cost, such a firm may therefore be able to produce and market new innovations. These arguments suggest a positive relationship between firm size and a firm's focusing on innovation and risk taking.

Small Firms

Research does not fully support the above large-firm hypothesis.

Laboratories of large firms may become overorganized (Scherer & Ross, 1990). When too many people are involved in an R&D project, they may spend a disproportionate amount of their time in writing memoranda to each other at the expense of creative endeavor. Also, it is commonplace to find the most able researcher in a large laboratory giving up work at the bench, and instead devoting nearly all his/her time to supervising others. Furthermore, in large firms, decisions may have to filter through an elaborate organizational hierarchy. This system of bureaucratic controls frequently drives creative individuals, who are not able to get ideas approved by higher management, out of large firms to go it alone with their own ventures (Robert, 1968; Scherer & Ross, 1990). On the other hand, small firms may be more adept at risk taking because their risky innovation projects generally are made by a group of people who know each other well.

In light of these arguments, the following competing hypotheses are proposed for testing the relationship between firm size and entry timing.

Hypothesis 11a: The larger the size, the earlier the firm will enter an emerging product market.

Hypothesis 11b: The smaller the size, the earlier the firm will enter an emerging product market.

Control Variables: Other Influences on Entry Timing

Other factors may also affect entry timing. First, whether the entrant is an industry incumbent or newcomer may influence its entry timing (Mitchell, 1989, 1991). Industry incumbents are those firms which have already had experience in the industry to which the emerging product belongs. The possible costs of cutting into the sales of existing products create disincentive for incumbent firms to enter an emerging product market earlier (Tushman & Anderson, 1986; Mitchell, 1989). Additionally, incumbents may also be constrained by structural inertia, that is, the tight links between several organizations may delay action by each other (Granovetter, 1973).

It is argued that capacity of task environment may affect firm's behavior (Dess & Beard, 1984). Market size can be an indicator of environmental carrying capacity (Brittain, 1994). Thus, it is reasonable to expect that large market size provides incentives attracting firms to enter earlier (Mitchell, 1989).

Entry timing into an emerging product market may also be influenced by the number of potential rivals (Mitchell, 1989). Competitive pressure is likely to cause earlier product development. Research suggests that, in general, the greater the number of competitors, the greater is the incentive to act early rather than wait (Wernerfelt & Karnani, 1987).

Entry timing may also be explained by industry-specific behavior. For example, for some industries, the second-movers may enter the emerging markets several years after they were created by first-movers (e.g., 5 years for disposable diapers and 3 years for smoke detectors), while for some other industries, the second-movers or even the third and fourth entrants enter the emerging markets within a few months, right after first-movers introduced the first products (e.g., camcorders). This phenomenon may not be explained by the factors discussed above. In this dissertation, it is classified into the influence of industry-specific entry behavior, which is different across industries and is expected to affect individual firm's entry timing.

CHAPTER 4

RESEARCH METHODS

Design of Investigation

Choice of New Product Categories for Study

A multi-industry approach has been employed in previous entry timing studies (e.g., Robinson, 1988; Robinson & Fornell, 1985; Mitchell, 1989). Following the design of these previous studies, this research examines the effects of organizational characteristics on entry timing based on the longitudinal entry events of ten emerging product markets. In this research, the selection of product categories industries was based on the following criteria:

- (1) Each of the product categories has been viewed by the public as a major innovation which created a significant emerging market. This criterion can be operationalized by the new product category that has created a new category in Business Periodical Index (e.g., the digital watch and the electronic cash register) or a new seven-digit SIC code in F&S Predicasts (e.g., the personal computer and the facsimile machine).
- (2) Each of the product categories must have a reasonable number of entrants on which data for all variables investigated can be gathered. After a preliminary review of possible data sources, the data for American public

firms were considered to be much more accessible, particularly for financial statements and TMT biographical data. Foreign and private firms might have the problem of incomplete data sets. Therefore, the selected product categories should include a reasonable number of American public firms to constitute the sample of the research. Each product category selected for analysis should include at least five American publicly owned firms.

(3) Only products with major long-term growth markets were considered. Product categories that were marketed locally and/or with insignificant sales were eliminated.

(4) To ensure data availability, only product categories introduced after 1960 were considered. This criterion was included because two critical data sources for this research, Business Periodical Index and Dun's Million Dollar Directory were first published in 1958 and 1959 respectively.

(5) The first-mover of the product and the first year it was introduced to the market can be clearly identified. It is critical to identify the first introduction year because it plays the role of base when calculating each entrant's waiting time to entry, the dependent variable of this dissertation.

Based on these criteria, ten products were selected. Table 2 presents these product categories, and the first-mover, first introduction year, number of entrants with complete data set, and average waiting time to entry for each of the product categories.

Table 2 Selected Products, First-Movers and First Introduction Year

Product	First-Mover	Number of Entrants with Complete Data	Average Waiting Time to Entry (in years)	First Introduction Year
Disposable Diaper	Procter & Gamble	9	9.7	1961
Facsimile Machine	Xerox	15	8.2	1964
Microwave Oven	Raytheon - Amana	13	7.7	1966
Electronic Cash Register	Ricca Data	17	4.7	1968
Digital Watch	Hamilton Watch	9	4.2	1971
Hand-Held Calculator	Bowmar Instruments	14	2.9	1971
Smoke Detector	General Electric	11	4.4	1972
Video Game	Magnavox	8	4.3	1973
Personal Computer	MITS	16	7.6	1975
Camcorder	Eastman Kodak	6	1.6	1984

Data Collection for Entrants and Entry Timing

The second step of data collection was to identify each entrant and its entry timing for each of the selected product categories. The method utilized here has been labelled structured content analysis (Jauch, Osborn & Martin, 1980). This methodology has been previously employed to analyze competitive interaction (Smith, Grimm & Gannon, 1992). This research extended its utilization to identifying entrants and entry timing for each of the product categories selected for analysis. All the F&S Predicasts and Business Periodical Index issues from the first year a product was introduced through the end of 1993 were searched by SIC code or appropriate category for headline summaries indicating firms' entry timing. The name of the entrant, the article headline, the publication, the publication date, and the page number were noted. This procedure was completed for all ten product categories and approximately 500 articles were identified. These articles were widely scattered in numerous general business periodicals (e.g., Business Week, Wall Street Journal, Fortune, Forbes, Merchandising Week, Merchandising Month, Advertising Age, etc.) and industry specific periodicals (e.g., Electronic News, Electronics, Business Automation, Computerworld, Datamation, etc.). An entry was identified by a key word search of each of the articles. Key words included "enter," "entry," "move into," etc. When reading each of these articles, the names of the reported entrants were noted as well as the time of entry.

Data Collection for Top Management Team Demographic Characteristics

TMT data collection started with identifying team members. The issues of Standard & Poor's Register of Corporations, Directors and Executives for the years 1961, 1964, 1966, 1968, 1971, 1972, 1973, 1975 and 1984 were used to obtain the list of TMT for each entrant of each of the product categories. Each list included top managers' names and positions. Because no single biographical data source could completely meet the requirements of this research, an aggregate approach become necessary for gathering data of TMT demographic characteristics. TMT data for each entrant, including age, organizational tenure and educational levels, were obtained by aggregating the information provided by Standard & Poor's Register of Corporations, Directors and Executives, Dun & Bradstreet's Reference Book of Corporate Managements (first issue published in 1967), and Who's Who in Finance and Industry for the years 1961, 1964, 1966, 1968, 1971, 1972, 1973, 1975 and 1984. For example, Standard & Poor's Register of Corporations, Directors and Executives includes more complete lists of TMT, but it does not provide complete information about organizational tenure. Therefore, supplementary collection for TMT organizational tenure data can be done by using Dun & Bradstreet's Reference Book of Corporate Managements and Who's Who in Finance and Industry.

Data Collection for Other Organizational Characteristics

Calculating return on sale and debt ratio required the information from consolidated income statement and balance sheet. Firms' financial statements were collected from Moody's Industrial Manual and Public Utility Manual for relevant years. Each entrant's data for SIC codes, number of employees, annual revenue, and board of directors were gathered from Standard & Poor's Register of Corporations, Directors and Executives for the corresponding years. Dun's Million Dollar Directory offered a source for double-checking.

Data Collection for Control Variables

Annual data for each of the ten industries were collected for industry sales. The primary sources for industry sales included statistical publications and general business and industry specific periodicals. For personal computer, microwave oven, smoke detector, video game, and disposable diapers, data could be collected from annual issues of Survey of Current Business (Office of Business Economics) and Standard & Poor's Predicasts BaseBook. For facsimile machine, electronic cash register, digital watch, hand-held calculator, and camcorders, data were collected from various business periodicals. The number of industry incumbents for each of the products was gathered from the Dun's Million Dollar Directory for relevant years. Dun's Million Dollar Directory provides a list of firms categorized

by four-digit SIC codes.

Operationalization and Measurement of Variables

Dependent Variable:

Entry Waiting Time. A timing variable records the number of years an entrant waits before entering an emerging product market (with the count starting at 1 for a firm that enters the first year). For example, the personal computer was first introduced in 1975 and IBM entered in 1981. Therefore, its entry waiting time is 6 years. The average waiting time to entry of the sample of this research is 5.76 years with a standard deviation of 4.06 years.

Independent Variables:

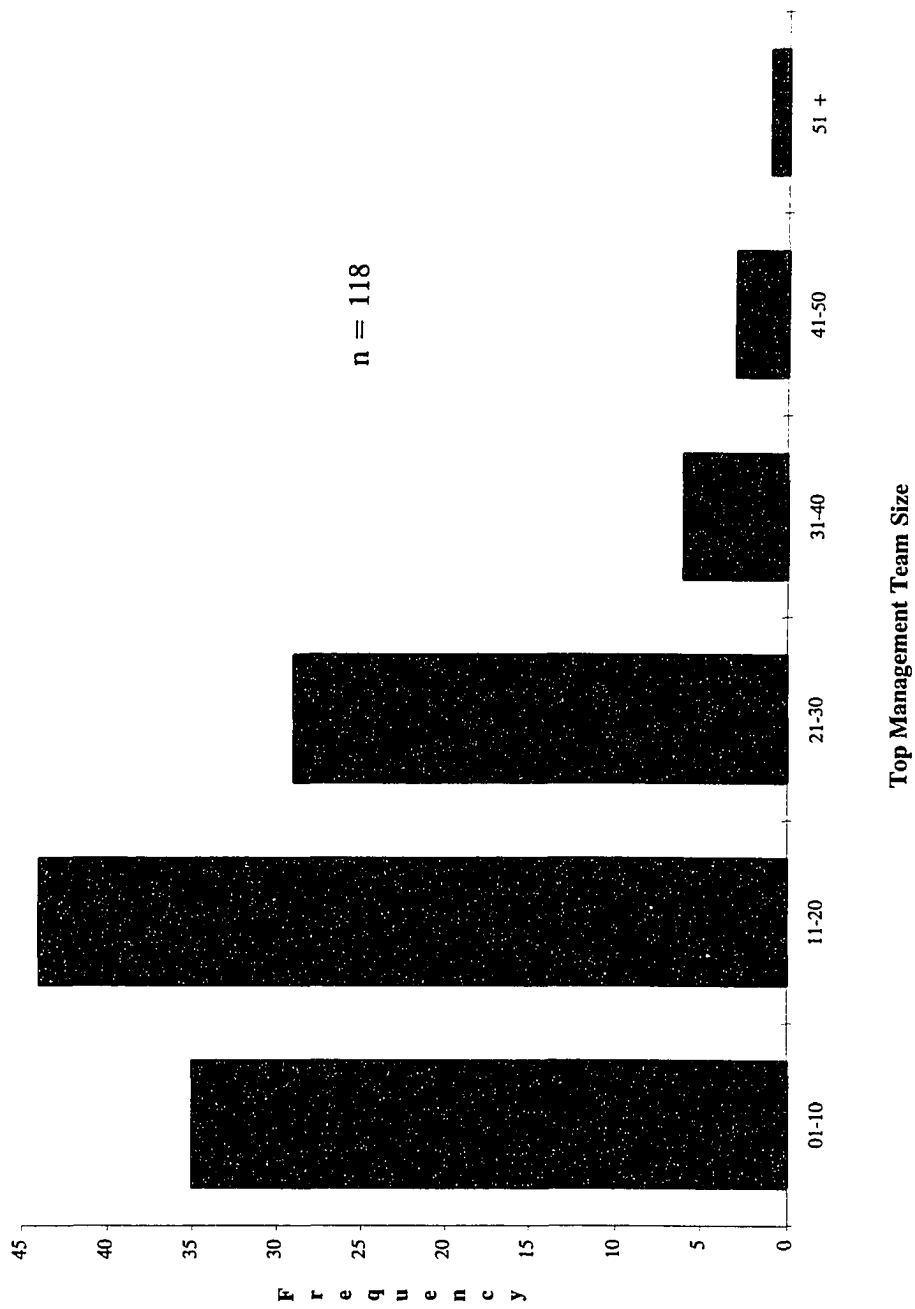
TMT Demographic Traits. Following O'Reilly and Flatt's (1989) study of effects of TMT demography on innovation, this research utilizes a definition of TMT to include the corporate officers listed in Standard & Poor's Register of Corporations, Directors and Executives, with title positions no lower than vice president. Three positions, including controller, general counsel and secretary, if not vice president, are included in TMT. Compared to previous studies, this dissertation uses a

broader definition to include more corporate managers in TMT for two reasons. First, because new product development often involves a bottom-up process (Burgelman & Sayles, 1986), it is reasonable to expect that corporate officers lower than "the two highest levels of management" (Wiersema & Bantel, 1993: p.492) may still have influences on new product related decisions. Second, new product development involves not only the efforts of R&D and marketing people, but also the supports/constraints from manufacturing, finance and legal departments (see Burgelman & Sayles, 1986: p.14). The size of top management team ranged from 3 to 52 individuals, with a mean of 17.06 and a standard deviation of 9.75. Figure 2 presents the distribution of the TMT size.

In this research, TMT traits include age, organizational tenure, and educational level. These TMT demographic traits were calculated by aggregating across all individuals within the TMT and taking the mean.

TMT Demographic Diversity. TMT demographic diversity refers here to the relative homogeneity/heterogeneity of TMT members in terms of functional area and levels of education. Diversity of TMT levels of education, on the interval scale data, was measured using the coefficient of variation (standard deviation divided by the mean). Allison's (1978) review of inequality measures indicates that the coefficient of variation, because it is a scale invariant measure, is to be preferred over the standard deviation or variance for interval level variables. For TMT functional area, a categorical variable, Blau's (1977) index of heterogeneity

Figure 2: The Distribution of Top Management Team Size



was applied. Let P_i is the proportion of the group's individuals in the i th category, then, Blau's Index of Heterogeneity is equal to $1 - \sum (P_i)^2$. This index has been used by previous studies (e.g., Bantel & Jackson, 1989; Lant, Milliken & Batra, 1992) to measure TMT heterogeneity in functional area. TMT functions were categorized into marketing, R&D, general management, finance and accounting, manufacturing and personnel and law.

Composition of Board. Board composition was operationalized as the ratio of inside members divided by board size (Beatty & Zajac, 1994). Inside directors are managers of the firm.

CEO-Board Chairperson Duality. This variable is binary. Following previous work, this research coded firms in which the CEO was also the chairperson of the board 1 and coded other firms 0 (Rechner & Dalton, 1991; Finkelstein & D'Aveni, 1994).

Levels of Diversification. Levels of diversification was operationalized by the number of four-digit SIC codes. Numerous researchers have effectively used a firm's SIC codes, which describe the types of businesses it competes in, to measure diversification posture (e.g., Montgomery, 1982; Finkelstein, 1992). Montgomery (1982) found that the product counts based on SIC codes yielded results that closely paralleled Rumelt's (1974) more intensive approach to assessing

levels of diversification.

Organizational Performance. In this research, performance was operationalized by return on sale (ROS). ROS was measured using net income divided by total sales.

Capital Structure: Debt Position. Debt position was measured by the ratio of total debt to total assets. This ratio captures the levels of debt in a firm's capital structure.

Boundary Spanning. Boundary spanning was operationalized by the ratio of TMT members who are more likely to engage boundary spanning activities total TMT size. As defined, boundary spanners are units or individuals operating on the peripheries of a firm and bringing in market and technical information from the environment (Smith, Grimm & Gannon, 1992). According to this definition, in this research, boundary spanners included the TMT members whose functions were categorized into marketing, R&D, regional executive, and product division.

Firm Size. Firm size was measured by the natural log of firm sales during the year before a product category emerged (deflated by the 1967 Producer Price Index).

Control Variables:

Incumbent Firm. To control for entry actions induced by experience with similar products, a dummy variable was coded as 1 if an entrant's four-digit SIC codes indicating its operations included the SIC code of the emerging product and 0 otherwise.

Number of Potential Rivals. The number of potential rivals can be represented by the number of industry incumbents (Mitchell, 1989). In this research, this variable was operationalized by the number of firms whose operations cover the two-digit SIC code of the emerging product.

Market Size. To control for market size incentives, this variable was measured by aggregate sales in the first five years of each of the product categories (deflated by the 1967 Producer Price Index).

Industry Specific Entry Behavior. This variable was measured by calculating the average waiting time to entry of the first one third entrants for each of the product categories.

Table 3 provides the descriptive statistics for variables used in the analysis

of waiting time to entry into emerging product markets.

Table 3
Descriptive Statistics for Variables Used in Analysis of Entry Timing^a

Variable	Mean	S.D.	Min.	Max.
Waiting time to entry	5.76	4.06	1	20
TMT age	51.97	3.73	42.25	63.67
TMT organizational tenure	18.33	6.88	2.20	32.59
TMT educational level	16.50	.97	13.33	18.53
Educational level diversity	.10	.05	.06	.30
Functional area diversity	.46	.26	.20	.80
Board composition	.41	.17	.09	1
Level of diversification	6.38	6.78	1	51
Boundary spanning	.35	.17	0	.7
Debt ratio	.48	.16	.08	.93
Firm performance (ROS)	.04	.06	-.42	.20
Firm size ^b	5.78	1.99	1.57	10.52
Number of potential rivals	2226.75	1133.64	609	3741
Market size ^b	6.39	1.19	4.78	8.06
Industry specific entry behavior	2.24	1.04	1.5	5.75
Frequencies			No	Yes
CEO duality			33	85
Incumbent firm			79	39

^an=118.

^bMeasured as natural log of millions of dollars, deflated by the 1967 Producer Price Index.

Data Analysis Methods

Accelerated Event-Time Analysis

To formulate a model of factors that influence the number of years it takes a contemplating firm to enter an emerging product market, this dissertation used the statistical framework of accelerated event-time analysis, an event-history approach (Kalbfleisch & Prentice, 1980; Tuma & Hannan, 1984). The advantage of event-history analysis is that it takes account of both the occurrence and timing of an event while estimating the effects of exogenous factors (Schoonhoven, Eisenhardt & Lyman, 1990). Analyses were performed using the LIFEREG and LIFETEST procedures in SAS (SAS Institute, 1990).

The accelerated event-time method assumes that the event times (in this dissertation research, entry waiting periods) are distributed according to a parametric baseline distribution that would hold if the value of all the independent variables were zero (Kalbfleisch & Prentice, 1980; Mitchell, 1989). The effect of the independent variables is to alter the rate at which an event occurs. A positive coefficient accelerates the baseline and a negative coefficient decelerates it. The additive logarithmic form of the model used in this research is expressed as

$$\text{Ln}T_i = a + BX_i + se_i \quad (1)$$

T_i is the observed event-time of the i th case; X_i is vector of coefficients associated with the independent variables. The errors vector, e_i , is distributed according to the assumed parametric distribution and is scaled by a variance-related parameter.

Accelerated event-time analysis requires that an underlying distribution be specified. Although many distributions are possible, the generalized gamma, a three-parameter distribution, is particularly flexible, allowing for both monotonic and nonmonotonic models. The generalized gamma collapses to two-parameter log-normal model when the gamma shape-parameter approaches to infinity, and a two-parameter Weibull model when gamma shape-parameter equal to 1.

The estimation procedure used was maximum likelihood estimation, which begins with the observed data and iteratively calculates a set of coefficient parameters that best fits the data. The likelihood ratio statistic was used to test the validity of the estimated model (Kalbfleisch & Prentice, 1980). This test statistic is:

$$R(B) = 2[\text{Ln}L(B') - \text{Ln}L(B_0)] \quad (2)$$

Where $L(B')$ is the likelihood for the model with the vector of estimated parameters B' and $L(B_0)$ is the likelihood for null hypothesis with all parameters equal to zero. This statistic is Chi-square distributed, with degrees of freedom equal to the number of independent variables.

Life-Table Analysis

One of the commonly used techniques for survival data analysis is the life-table method. Gehan (1969) provides the life-table method for estimating survivalship, density and hazard functions. The life-table method requires a fairly large number of observations, at least 30, so that survival times can be grouped into intervals (Lee, 1980).

The life-table estimates are computed by counting the numbers of observations that fall into each of the time intervals $[t_i, t_{i+1})$, $i=1,2,3,4,\dots, s$. Let n_i' be the number of units entering the interval $[t_i, t_{i+1})$, and let d_i be the number of events occurring in the interval. Let $b_i = t_{i+1} - t_i$ and let $n_i = n_i' - w_i/2$, where w_i is the number of units censored in the interval, and n_i is known as the effective sample size of the interval $[t_i, t_{i+1})$. Let t_{mi} denote the midpoint of interval $[t_i, t_{i+1})$.

The conditional probability of an event in i th interval is estimated by $q_i = d_i/n_i$. Let $p_i = 1 - q_i$, then, the estimates of survival function at t_i , density function at t_{mi} , and hazard function at t_{mi} are

$$S(t_i) = \prod_{j=1}^i (1-q_j) \quad (3)$$

$$f(t_{mi}) = S(t_i)q_i/b_i \quad (4)$$

$$h(t_{mi}) = 2q_i/[b_i(1 + p_i)] \quad (5)$$

CHAPTER 5

RESEARCH RESULTS

Estimation of Underlying Hazard Function

Prior to applying accelerated failure time analysis to test the research model, a hazard function must be selected (Kalbfleisch & Prentice, 1980; Tuma & Hannan, 1984). Figure 2 depicts a smoothed hazard function for the entry timing of entrants studied. The hazard function was estimated by life-table analysis which can be performed using LIFETEST procedure in SAS (SAS Institute, 1990).

The entry-time plot reveals that the distribution of entry events across waiting times is approximately monotonic and exhibits an increasing tendency. Based on this graphic analysis, a Weibull distribution may best model the data. To determine if the underlying distribution matches the empirical form, the comparison between the Weibull and lognormal distribution was made by testing the absolute difference of their log-likelihoods multiplied by two, a difference having a Chi-square distribution (Kalbfleisch & Prentice, 1980). The Weibull distribution yielded a better fit than the lognormal distribution ($\chi^2 = 13.04$, $p < .01$). As a result, the Weibull distribution was chosen for the model.

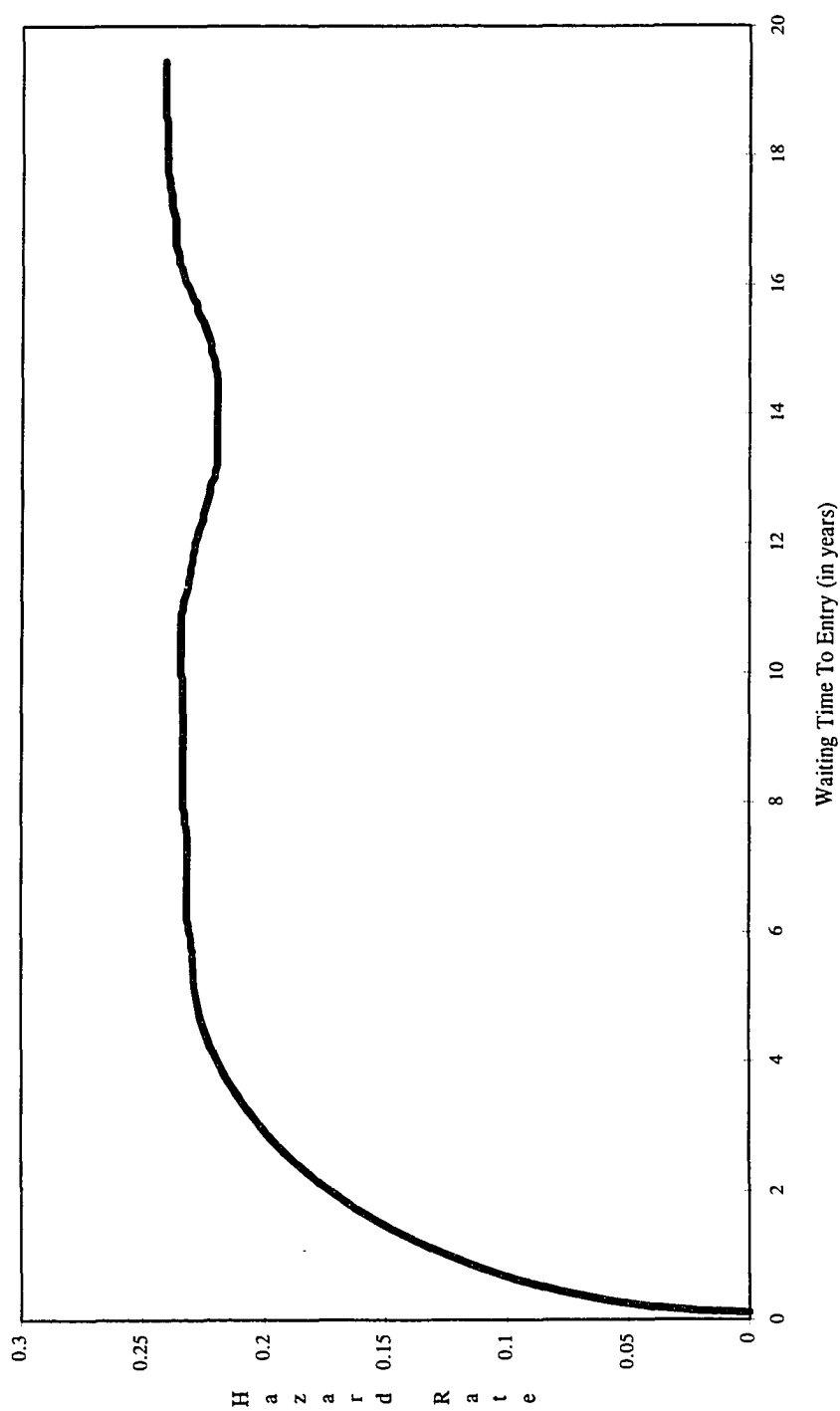


Figure 3 Hazard Function Estimation

Results for the Hypothesized relationships Between Organizational Characteristics and Entry Timing

Table 4 shows the intercorrelations for the independent variables. Although a number of variables are significantly correlated, correlation magnitudes do not indicate cause for concern about multicollinearity. However, for the high intercorrelations between TMT tenure and TMT age(.56), and market size and number of potential rivals (.68), attempts at reestimating the models after removing one independent variable at a time from the equations did not significantly affect the results. Since theory indicates that each of the independent variables should affect a firm's entry timing, the decision was made to leave all variables in the models.

Table 5 contains the maximum likelihood estimates of five models predicting a firm's waiting time to entry into an emerging product market. Model 1 estimates a model with 16 covariates, and it contains variables that directly test all the hypotheses as well as including the control variables. Each of the remaining models in Table 4 (model 2 to 5) are similar to model 1, but a blocked analysis approach was used to examine the extent to which blocks of independent variables contribute to fit the data (Shoonhoven, Eisenhardt, Lyman, 1990). This is done by eliminating blocks from the model and examining whether the log-likelihood ratio statistic is significantly better when a given block of variables is

Table 4
Intercorrelations for Independent Variables^a

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1.TMT Age															
2.TMT Education	-.18														
3.TMT Tenure	.56	-.17													
4.Educational diversity	.27	-.46	.03												
5.Functional diversity	.00	.29	.23	-.11											
6.Board composition	.02	-.22	-.03	.27	.05										
7.CEO duality	.16	-.10	.07	.06	-.02	.19									
8.Levels of diversification	.14	.25	-.08	-.04	-.09	-.13	-.01								
9.Boundary spanning	-.18	.23	.05	-.16	.25	-.13	-.15	.05							
10.Performance: ROS	.10	.15	.26	-.02	.08	.04	-.08	-.12	.05						
11.Debt ratio	.03	-.01	-.28	.11	-.10	-.03	.07	.15	-.02	-.45					
12.Firm size	.29	.36	.33	-.11	.29	-.30	.14	.28	.26	.20	.11				
13.Potential rivals	.21	.30	-.01	-.09	.12	-.19	.07	.30	.05	.11	.21	.34			
14.Incumbent firm	-.12	-.02	-.19	-.14	-.01	.00	-.02	.35	-.01	-.16	.07	-.06	.24		
15.Market size	.15	.30	-.08	-.11	.19	-.25	.02	.28	.07	-.12	.31	.21	.68	.22	
16.Industry specific entry behavior	-.08	.07	.07	-.22	.02	.14	.03	-.10	-.05	.15	.33	.01	-.29	-.06	-.35

^a n = 118. Coefficients greater than .17 are significant at p < .05.

Table 5
Maximum Likelihood Estimates Predicting Waiting Time to Entry^{a,b,c}

Variables	Model 1		Model 2		Model 3	
TMT Traits:						
Age	.003	(.065)	.013	(.064)	.008	(.065)
Educational levels	.134 [†]	(.068)	.130 [†]	(.069)	.143*	(.066)
Organizational tenure	.020	(.067)	.015	(.066)	.019	(.067)
TMT Diversity:						
Educational levels	.246*	(.102)	.265**	(.098)	.247*	(.102)
Functional area	-.007	(.061)	.001	(.057)	-.018	(.057)
Corporate governance:						
Board composition	.150	(.404)			.048	(.074)
CEO duality	.024	(.106)			.012	(.104)
Level of diversification	.029	(.058)	.032	(.056)		
Boundary spanning	-.123 [†]	(.341)	-.121 [†]	(.065)	-.124 [†]	(.065)
Financial situation:						
Firm performance	.204**	(.065)	.208**	(.065)	.198**	(.065)
Debt ratio	.141*	(.070)	.147*	(.070)	.140*	(.071)
Firm size	-.186*	(.094)	-.203*	(.039)	-.168 [†]	(.089)
Control variables:						
Number of potential rivals	.035	(.075)	.039	(.075)	.034	(.075)
Incumbent firm	-.039	(.117)	-.048	(.116)	-.021	(.111)
Market size	-.261***	(.059)	-.273***	(.074)	-.255***	(.075)
Industry specific entry behavior	.292***	(.076)	.309***	(.054)	.300***	(.057)
Intercept	1.107***	(.168)	1.096***	(.147)	1.105***	(.168)
Weibull scale parameter	.479	(.035)	.479	(.035)	.480	(.035)
Model log-likelihood	-99.128		-99.339		-99.256	
Log-likelihood ratio statistic	59.744		59.324		59.488	
Degrees of freedom	16		14		15	
Model significance	p < .001		p < .001		p < .001	

^aStandard errors are in parentheses. [†] p < .10, *p < .05, **p < .01, ***p < .001 (two-tailed tests).

^bn = 118.

^cThe independent variables are standardized to mean=0 and standard deviation=1.

Table 5 (continued)
Maximum Likelihood Estimates Predicting Waiting Time to Entry^{a,b,c}

Variables	Model 4		Model 5	
TMT Traits:				
Age	.019	(.063)	-.004	(.065)
Educational levels	.140*	(.066)	.117 [†]	(.068)
Organizational tenure	.013	(.066)	-.003	(.065)
TMT Diversity:				
Educational levels	.268**	(.098)	.285**	(.102)
Functional area	-.008	(.054)	-.048	(.056)
Corporate governance:				
Board composition				
CEO duality				
Level of diversification				
Boundary spanning	-.122 [†]	(.065)	-.148 [†]	(.068)
Financial situation:				
Firm performance	.202**	(.065)		
Debt ratio	.146*	(.070)		
Firm size	-.189*	(.081)		
Control variables:				
Number of potential rivals	.037	(.075)	.055	(.075)
Incumbent firm	-.026	(.110)	-.085	(.115)
Market size	-.268***	(.073)	-.255***	(.073)
Industry specific entry behavior	.312***	(.054)	.315***	(.058)
Intercept	1.080***	(.144)	1.050***	(.153)
Weibull scale parameter	.480	(.035)	.508	(.036)
Model log-likelihood	-99.503		-104.465	
Log-likelihood ratio statistic	58.998		49.070	
Degrees of freedom	13		10	
Model significance	p < .001		p < .001	

^aStandard errors are in parentheses. [†] p < .10, *p < .05, **p < .01, ***p < .001 (two-tailed tests).

^bn = 118.

^cThe independent variables are standardized to mean=0 and standard deviation=1.

included. For performing the blocked analysis, the independent variables were categorized into seven blocks: TMT traits (age, educational levels and organizational tenure), TMT diversity (educational levels and functional area), corporate governance (board composition and CEO duality), levels of diversification, boundary spanning, financial status (organizational performance and capital structure) and firm size.

Overall, Model 1 is a significant improvement over a model with no covariate -- the likelihood ratio of 59.744 ($p < .001$) indicates that the model is a good fit to the data. Regarding tests of specific hypotheses, Table 5 indicates that TMT educational level has a statistically significant effect on extending the entry time, however, the effect is the opposite of the predicted direction. For firms with more educated TMTs, there is an increase in the waiting time to entry into emerging markets. The unexpected results will be discussed in the next chapter. The effects of both TMT age and TMT organizational tenure on entry timing are not statistically significant.

The results for the effect of TMT diversity in educational levels on waiting time to entry are statistically significant and support the homogeneity hypothesis. That is, the greater the TMT homogeneity in educational levels, the earlier the firm enters an emerging market. The results for the effect of TMT functional area diversity on entry timing are not statistically significant.

Neither the board composition, nor the CEO duality has a significant impact on waiting time to entry. The results may be attributable to the agency and organic structure problems that will be discussed in detail in the next chapter.

Results for the effect of levels of diversification on entry waiting time are not statistically significant. Boundary spanning activities affect entry timing in the expected direction and the results are statistically significant. That is, the greater the boundary spanning activities, the shorter the entry waiting time.

Firm performance has a significant effect on waiting time to entry. The results support the argument that for firms that perform well, rather than badly, there is an increase in the waiting time to enter into emerging product markets. Debt position also has a significant effect on entry timing, and in the predicted direction. The higher the debt position in capital structure, the longer the waiting time to entry into an emerging product market.

Results for the impact of firm size on waiting time are significant. The results show a support for the Schumpeterian hypothesis; that is, larger firms are more innovative and can afford the costs associated with failure and, in turn, enter emerging product markets earlier. Table 6 summarizes the results of the tests of specific hypotheses.

Table 6
Summary of Results for Tests of Hypotheses

Hypothesis Number	Independent Variables	Expected Sign	Conclusion from Hypothesis Test
1	TMT Age	(+)	Not Significant
2	TMT Organizational Tenure	(+)	Not Significant
3	TMT educational Level	(-)	Significant -- Contradict
4	TMT Diversity	(c) ^a	Significant (Homogeneity -- > Early Entry)
5	Proportion of Inside Directors	(-)	Not Significant
6	CEO Duality	(-)	Not Significant
7	Level of Diversification	(+)	Not Significant
8	Firm Performance	(c) ^a	Significant (Low Performance -- > Early Entry)
9	Boundary Spanning	(-)	Significant
10	Debt Ratio	(+)	Significant
11	Firm Size	(c) ^a	Significant (Large Size -- > Early Entry)

^a(c) denotes competing hypotheses.

The results for the effects of the control variables on waiting time to entry also reveal valuable information. Market size, which was measured by aggregate five-year industry sales, significantly affects the entry waiting time of individual entrants. Larger market size provides incentive attracting firms to enter emerging markets earlier. Neither the number of potential rivals nor the incumbent firm has a significant impact on individual firm's waiting time. The last control variable expected to influence waiting time is industry-specific entry behavior. The results show a significant effect on entry timing, in the predicted direction.

After presenting the results for the full model, the next step is to further examine the corporate governance and level of diversification blocks of covariates to assess their relative contributions to the fit. Since no individual variable in either block had a significant influence on entry timing, it is necessary to explore the extent to which inclusion of these variables contributes to knowledge about timing of entry into emerging markets (Shoonhoven, Eisenhardt & Lyman, 1990).

In Table 5, model 2 through 5 are hierarchically related to model 1. Each was estimated by sequentially eliminating blocks of covariates from the full model. The procedure was to drop blocks of covariates and compare the relative fit of each subsequent model. The purpose here was to find the best-fitting model and thereby increase understanding of the relevant theoretical factors. Model 2

through 5 are all significant improvements over the model with no covariate. For all four models the log-likelihood ratio statistic is significant at the level of $p < .001$.

As in model 1, none of the variables in corporate governance block had a significant effect on waiting time to entry. In model 2, this block, including board composition and CEO duality, was eliminated and the log-likelihood ratio statistic obtained in the reduced model 2 was compared with that obtained in model 1. The comparison was made by calculating the difference in the log-likelihood ratio statistics for the two models and the difference in degree of freedom. The results show that the difference statistic, $\chi^2 = .421$ with 2 degrees of freedom, is not statistically significant ($p < .75$). That is, the exclusion of the corporate governance variables from the larger model provides no less in fit. Therefore, it can be concluded that the main effects of the board related corporate governance variables taken as a block do not add significantly to our knowledge about the waiting time to entry into emerging product markets.

Model 1 also showed that the levels of diversification had no significant effects on waiting time to entry. Model 3 drops out this single-variable block. The log-likelihood ratio statistics of model 1 and model 3 are compared, and a difference value of $\chi^2 = .256$ with 1 degrees of freedom is obtained. Since the chi-square value is not significant ($p < .75$), there is no statistically significant

difference between these two models. Thus, it can be concluded that levels of diversification have no substantive impact on waiting time to entry into emerging product markets.

In model 4 these two block of covariates, corporate governance and levels of diversification, are dropped out from the full model. When the log-likelihood ratio statistics are compared, the difference value of $\chi^2=.749$ with 3 degrees of freedom is also not statistically significant ($p<.75$). Therefore, it can be concluded that compared to models 1, 2, and 3, model 4 with fewer covariates is preferred as more parsimonious.

Last, in model 5, only TMT related variables, including TMT age, TMT organizational tenure, TMT educational levels, TMT educational level heterogeneity, TMT functional area heterogeneity, and boundary spanning (measured by the ratio of TMT members who are more likely to engage in boundary spanning activities to TMT size) are retained, while other independent variables are eliminated from the model. The log-likelihood ratio statistics of model 1 and model 5 are compared, and a difference value of $\chi^2=10.673$ with 6 degrees of freedom is obtained. The chi-square value is around the margin of statistical significance ($p=.1$ where $\chi^2=10.645$). However, when the log-likelihood statistics of model 4 and model 5 are compared, the difference value of $\chi^2=9.924$ with 3 degrees of freedom is statistically significant ($p<.025$).

Therefore, it can be concluded that model 4 is preferred because the additional variables, including performance, debt position, and firm size, provide a significant improvement in fitting the data. However, the results indicate that the "managerial model" has been a strong competitor in predicting entry timing.

CHAPTER 6

RESEARCH RESULT DISCUSSION AND CONCLUSIONS

Discussion

This dissertation develops hypotheses and empirically investigates the relationships between organizational characteristics and entry timing. This study responds to the suggestions for endogenous entry timing models by previous investigators (Lieberman & Montgomery, 1988; Moore, Bounding & Goodstein, 1991). As Lieberman and Montgomery (1988:p.52) say, "most theoretical studies have begun by assuming some initial asymmetry, thereby sidestepping the endogeneity issue." For empirical research, they add, "the endogeneity issue has been ignored in most empirical studies as well as in theoretical work" (p.53). Specifically, previous research suggests that models with endogenous structure are necessary for investigating the linkage between firm attributes and entry timing (Lieberman & Montgomery, 1988; Moore, Boulding & Goodstein, 1991). This study supports the general argument of the endogenous entry timing research that not all firms have the same entry costs and skills; therefore, a firm's entry timing decision is partially attributable to factors internal to the organization.

Theoretically and empirically, this research adds to our understanding of the endogeneity of entry timing decisions. That is, entry timing, like many other

strategic decisions, can be explained in part by organizational characteristics. The estimates of the effects of organizational characteristics on entry timing as hypothesized in this research, are robust. This study finds that the firms that enter emerging product markets earlier have less educated and more homogeneous TMTs, lower performance, greater boundary spanning activities, lower ratio of debt to total capital, and are of larger size.

This study introduces and empirically tests the "strategic choice" view that the top management team influences a firm's timing of entry into an emerging product market (Child, 1972). While prior research has examined the TMT - strategy type linkage, a focus on a firm's entry timing, as investigated in this research, has not been done previously. Although the model with only TMT variables does not come out as the best model in fitting the data, the results show that it still is usable in predicting firms' entry timing. The results of this research provide support for the "upper echelons" perspective of Hambrick and Mason (1984). TMT characteristics serve as an important basis by which individuals cognitive base can be measured; cognitive bases, in turn, combine to create certain team ability and knowledge, resulting in patterns of decision outcomes (Wiersema & Bantel, 1992).

TMT homogeneity, rather than heterogeneity, was found to speed firms' entry into emerging markets. This finding implies that social integration associated

with team homogeneity is more appropriate for explaining firms' early entry than diverse sources and interpretation of information related to team heterogeneity (O'Reilly & Flatt, 1989; Wiersema & Bantel, 1992; Smith et al., 1994). The process of innovation is likely to generate conflict and confusion (O'Reilly & Flatt, 1989). To overcome these problems requires managerial characteristics of teamwork, flexibility, and integration. Previous research suggests that successful innovation is partly a function of effective communication and integration among organizational units provided by top management (Shrivastava & Souder, 1985). Present results support the view that the group homogeneity is associated with greater social integration, and thereby permits more effective communication across organizational functions, and provides quicker adoption of innovations. This effect can be illustrated by individual entrants in the sample. Xerox (fax machine), General Electric (microwave oven), Hamilton Watch (digital watch), General Electric, Emhart, and Honeywell (smoke detector), with higher TMT homogeneity in educational level, entered the emerging product markets early. On the other hand, Johnson & Johnson (disposable diaper), IBM (fax machine), Sunbeam, Maytag, Welbilt (microwave oven), Gillette (digital watch), Beckman (hand-held calculator), Mattel, and General Instruments (video game), with higher heterogeneity in TMT educational level, entered the product markets late.

An unexpected finding was that lower, rather than higher, top management team educational levels are significantly associated with early entry timing.

Educational levels have been linked to information search activities (Smith, Grimm & Gannon, 1992; Wiersema & Bantel, 1992). According to such research, more educated managers tend to be more exhaustive in information searching and this emphasis results in greater richness from which to make a decision. This may imply that more educated managers are more dependent on information in decision making processes. However, as discussed, early entrants confront more uncertain technical and market situations where information is less likely to be available (Galbraith, 1973; Fredrickson & Mitchell, 1984). Thus, it seems reasonable to expect that the TMT of an early entrant relies on less information when making an entry decision.

Furthermore, using a great quantity of information in decision making often represents a comprehensive process (Fredrickson & Mitchell, 1984). Such a decision process involves "carefully weighing the costs and risks of various consequences" (Fredrickson & Mitchell, 1984:p.402). However, this may frustrate organizational entrepreneurship which usually is observed in the organizations that can offer flexible financial support for risky projects (Quinn, 1979). Previous entrepreneurship research found that entrepreneurs with higher educational levels typically have lower performance than less educated entrepreneurs (Roberts, 1968; Douglass, 1976). For such results, Roberts (1968:p.263) concludes that highly educated entrepreneurs are typically lower rated performers because "their general temperament, attitude, and orientation are usually out of line with those needed for

successful entrepreneurship." The above arguments provide us potential explanations for the relationship between higher TMT educational levels and late entry. This effect can be demonstrated by individual entrants. Magnavox (fax machine), Hamilton Watch (digital watch), and Tandy (personal computer), with less educated TMTs, entered the emerging product markets earlier. However, Beckman (hand-held calculator), Texas Instruments (video game), ITT, Sperry, and Honeywell (personal computer), with more educated TMTs, entered the product markets later.

Boundary spanning, as measured by the ratio of the TMT members engaging in boundary spanning activities to the TMT size, is found to be related to early entry. This finding is consistent with the position of Miles and Snow (1978) that the dominant coalition of a prospector, which is characterized as the firm seeking always to be the first among the developers of new products, centers around the marketing and R&D functions. Further, the results partially support the work of Smith, Grimm and Gannon (1992). They found that firms with more boundary spanners in their TMTs compete more aggressively in markets, for example, through quick response to competitors' actions. The relationship between greater boundary spanning and early entry can be illustrated by General Electric, Tappan (microwave oven), Texas Instruments, Singer (hand-held calculator), General Electric (smoke detector), Kodak (camcorder). On the other hand, the effects of less boundary spanning on late entry can be illustrated by Johnson &

Johnson, Riegel Textile (disposable diapers), Gillette (digital watch and hand-held calculator), and PRIME Computer (personal computer).

A further analysis was conducted to understand whether R&D related TMT members have impact on firms' entry timing. To explore this, the number of TMT members whose functions are related to R&D and technology was divided by TMT size. This ratio was added to models 1, 4, and 5 as a new variable. The results were not statistically significant. A possible explanation may be that the R&D function of the firms studied has been relegated to product divisions. Therefore, the R&D related TMT members do not reflect a firm's emphasis on R&D effort.

The findings on the effects of firm performance are of particular interest. A relationship was found here between low performance and early entry. This result is consistent with Bowman's (1980, 1982) argument that troubled firms take more risk, as well as with Mann and March's (1978) findings that financial adversity stimulates innovation. This result, on the other hand, implies that although better performance may generate more organizational slack which allows firms the ability to absorb the costs associated with innovation and risk taking (Rosner, 1968; Singh, 1986; Damanpour, 1987), it seems that better performers are more committed to the status quo (Hambrick, Geletkanycz & Fredrickson, 1993). That is, change is less likely to be considered when firms perform well.

The case example of Eastman Kodak's pioneering the camcorder market illustrates the effect that this research observed statistically. Kodak introduced the world's first commercial camcorders in 1984, and this pioneering was described as "a risky leap" (Business Week, 1984, January 16:p.92). Kodak's risky action could be attributable to a combination of traumatic events in 1982 and 1983. During this period, Kodak had been hit hard by growing competition from Fuji Film, a slowdown in sales of various product lines, and a decreased market share in instant cameras. As a result, operating profits in 1982 dropped almost 10% despite a 5% revenue gain to 10.8 billion. Most analysts were projecting an even greater drop in 1983 earnings. These adverse situations forced Kodak to take the highly risky action. Except Kodak, other first-movers, including Raytheon-Amana (microwave oven), Hamilton Watch (digital watch), Bowmar (hand-held calculator), and Magnavox (video game), reported a low return on sale prior to the introduction of the products.

Capital structure with lower debt position was found to quicken a firm's entry, as expected. This finding implies that debt holders impose strict constraints on firms' risky innovation activities. These results support previous theoretical arguments that different sources of capital will influence a firm's commitment to innovation and attitude toward risk differently (Leland & Pyle, 1977; Long & Ravenscraft, 1993). This finding is consistent with the asymmetric information and moral hazard arguments of the agency theory. As has been argued, such

perspectives suggest that debt financing is inappropriate for product innovation projects. These effects can further be illustrated by individual entrants. For example, highly leveraged firms, including Mattel (video game), ITT, Honeywell, and PRIME Computer (personal computer), entered the emerging markets later. On the other, Procter & Gamble (disposable diapers), Tappan (microwave oven), Hewlett Packard, Texas Instruments (hand-held calculator), and Kodak (camcorder) had comparatively low debt ratios, and entered the emerging product markets earlier.

The results for the effect of firm size on entry timing are also worth noting. The larger firms, rather than smaller firms, enter emerging product markets earlier. This finding does not agree with the proposition of Wernerfelt and Karnani (1987) that larger firms are better able to enter later and yet snatch the leadership position for earlier entrants. On the contrary, this research supports the Schumpeterian hypothesis that economies of scale in R&D activities allow large firms to be more efficient in producing innovative output for a lower investment (Schumpeter, 1961).

However, when individual firms of the sample of this research are further examined, the results are worth noting. Large firms (e.g., the firm's annual revenue in excess of 1 billion dollars, after deflated by the 1967 Producer Price Index) do not constitute the majority of the first-movers of the product categories .

studied. More specifically, only Procter & Gamble (disposable diaper), Raytheon (microwave oven), General Electric (smoke detector), and Eastman Kodak (camcorder) are large firms. However, if the firms that enter emerging product markets within two years after they emerged are classified as fast followers (in this research, the average waiting time to entry is 5.76 years), many large firms are found to be fast followers, such as Singer, Litton Industries, Texas Instruments (hand-held calculator), General Electric (microwave oven), Singer, RCA (electronic cash register), General Electric, and RCA (camcorder).

No support for the effects of both board related corporate governance variables, board composition and CEO duality, was found in this study. The lack of significant finding for board composition may be attributable to the preference of high technology investors. Nine out of ten product categories studied in this research belong to electronic, computer and communication industries. These industries have been classified into high technology industries. Previous research theorizes that the interests of inside directors are aligned with those of management while the interests of outside directors are aligned with stockholders (Williamson, 1975; Hill & Snell, 1988; Baysinger & Hoskisson, 1990; Smart & Hitt, 1994). Most stockholders invest in a diversified portfolio of stocks in order to eliminate unsystematic risk. So long as their portfolio eliminates unsystematic risk, then most investors are prepared to hold some high-risk/high-return stocks (Hill & Snell, 1988). Within high technology industries, an emphasis upon competing

through research-based innovation is regarded as a high-risk/high-return strategy (Hay & Morris, 1979; Hill & Snell, 1988). Innovation strategies may be attractive to investors who hold a balanced portfolio, and who can switch out of a high-risk stock if the performance of the firm starts to slip. Therefore, outside directors, for reflecting the interests of stockholders, may urge managers to undertake more risky activities. As a result, although strategic controls may encourage managerial innovation and risk taking, in high technology industries, this impact may become indifferent because outside directors also force managers to focus on risky innovation investments.

The lack of significant findings for the effect of CEO duality can be explained by the organic structure argument (Burns & Stalker, 1961; Kanter, 1988). Burns and Stalker (1961) argue that organic structures, characterized by such things as flexibility in administrative relations, informality, and authority vested in situational expertise, can facilitate innovation. On the other hand, mechanistic structures, characterized by rigidity in administrative relations and formality, are argued to impede innovation. As hypothesized, CEO duality may enhance decisive actions avoid role and responsibility conflict, however, it may also create mechanistic structures. Therefore, these two contradictory forces may offset one another and, this makes the effect of CEO duality insignificant.

Finally, the lack of significant support for levels of diversification may

imply the inappropriateness for employing organizational control theory in explaining the relationship between levels of diversification and entry timing. Levels of diversification were hypothesized mainly based on the effects of strategic controls/financial controls on early/late entry. Control theory has been derived from the completeness of knowledge of cause/effect relations and clearness of standards of desirable performance (Thompson, 1967; Ouchi, 1977, 1978; Baysinger & Hoskisson, 1990; Snell, 1992). Snell (1992) found that corporate executives in more diversified firms have more complete knowledge of cause-effect relations than those in less diversified firms. If the theory is correct, then, a question arises: How diversified is diversified? Before this question is answered, it seems inappropriate to apply control theory to explain the relationship between levels of diversification and entry timing. In the present research, the firms studied were widely diversified. Levels of diversification, as measured by number of four-digit SIC codes, ranged from 1 to 51, with a mean of 6.38 and a standard deviation of 6.78. This sample included highly diversified firms (e.g., North American Rockwell, Singer, Sperry-Rand, North American Philips, Litton Industries) as well as single-business firms (e.g., Maytag, General Instruments, Data General, Wang Laboratory, Polaroid). Additionally, as firms increase their levels of complexity, corporate executives tend to retain control through centralized decision making (Child, 1973). It is suggested that these efforts may improve understanding of cause-effect relations as levels of diversification increase (Snell, 1992).

A Case Study: The Microwave Oven

The effects of TMT educational level, TMT educational level diversity, boundary spanning, firm performance, debt ratio, and firm size on entry timing are statistically supported by the data. In this section, the individual early and late entrants into microwave oven market are used to illustrate the effects that this research found statistically. Microwave ovens were first introduced by Raytheon-Amana in 1966. Table 7 depicts how these six organizational characteristics of the early entrants (entered before 1968) and late entrants (entered after 1976) into this emerging market fit the statistically significant findings.

The asterisk indicates the possession of a particular organizational characteristic by an individual firm. The relationship between educational levels and early entry can be found in Magic Chef and Geo Roper. On the other hand, Maytag, a firm with higher educational levels, entered the market 16 years after it was created. The linkage between TMT educational level homogeneity and early entry can be found in Tappan, General Electric, and Litton Industries. Later entrants, including Sunbeam, Maytag, Emerson, and Welbilt, had comparatively heterogeneous TMT. Greater boundary spanning speeded the entries of Tappan and General Electric, while less boundary spanning prolonged the entries of Maytag and Welbilt. Low performing firms, including Raytheon-Amana (first-mover), Tappan, and Geo Roper, entered the emerging market early. However,

Table 7
Organizational Characteristics of Early and Late
Entrants into Microwave Oven Market

Early Entrants

Firm	Low Education	Educational Homogeneity	Greater Boundary Spanning	Low Performance	Low Debt Ratio	Large Firm
Raytheon-Amana				*		*
Tappan		*	*	*	*	
General Electric		*	*			*
Magic Chef	*				*	
Geo Roper	*			*		
Litton Industries		*				*

Late Entrants

Firm	High Education	Educational Heterogeneity	Less Boundary Spanning	High Performance	High Debt Ratio	Small Firm
Sunbeam		*				
Maytag	*	*	*	*		*
Emerson		*			*	*
Welbilt		*	*			*

with high performance, Maytag entered the microwave oven market late. The effects of low debt position on early entry are found in Tappan and Magic Chef. However, Emerson whose debt ratio was high entered the market late. Finally, the effects of firm size on entry timing can also be observed in entrants into the microwave oven market. Raytheon, General Electric, and Litton Industries, with large size, entered the emerging market early. Smaller firms, including Sunbeam, Maytag, and Emerson, entered the market late.

Limitations of the Study

While this dissertation is innovative and contributes significantly to strategic management theory, it is not without its limitations. The most significant limitation of the study is that the sample was restricted to publicly owned firms, and the findings of this research may not be generalized to all firms in an industry. Differences can be found between publicly and privately owned firms. Public ownership can facilitate stockholder diversification, increase liquidity, and raise new capital more easily. However, Security and Exchange Commission (SEC) and public investors impose more strict constraints on the operations of publicly owned firms. They have to file annual and quarterly operating data with SEC, and such data will be then available to competitors. Also, there may be pressure on managers of publicly owned firms to produce annual earnings gains, even it would

be in the shareholders' best long-term interests to adopt a strategy that might penalize short-term earnings but lead to higher earnings in future years. Therefore, due to these differences, publicly owned firms may follow different entry strategies than privately owned firms. Future research should include privately owned firms, if possible.

Second, although a multi-industry approach is used in this research to investigate the relationships between organizational characteristics and entry timing, nine out of ten product categories selected for analysis in this study can be categorized into high-technology industries. The high-velocity environments of high-technology firms are described as "those environments in which rapid and discontinuous change occurs in demand, competition, technology, or regulation in such a way that information is often inaccurate, unavailable, or obsolete" (Eisenhardt & Bourgeois, 1990). The differences in environment between high-technology firms and medium- or low-technology firms may lead to different entry strategies across different industries. It would be useful for future research to include non-high-technology product categories, such as light beer, diet cola, or liquid laundry detergent. Furthermore, including service industries could provide a useful check of the generalizability of results found in the manufacturing industries.

Third, a possible limitation regards the nature of the survival data analysis employed in this research. The survival data analysis requires that a researcher

collects data on the predictor variables at the first measurement time, t_1 , and subsequently collects the dependent variables at a later time, t_2 . The interval between times t_1 and t_2 is called measurement window or data collection window (Singer & Willet, 1991; Morita, Lee & Mowday, 1993). Research suggests that if the selected window is too short, the process being observed may not have sufficient time to unfold and too little criterion variance may result (Morita, Lee & Mowday, 1993). On the other hand, if the window is too long, factors affecting the process that are not under study may arise. In this research, the longest waiting time to entry recorded is twenty years. Although this problem has not been addressed in previous entry timing research (e.g., Mitchell, 1989, 1991), it is reasonable to suspect that for such a long period of time there might be other factors influencing the entry process that are not under study. For dealing with this limitation, future research can include time-dependent covariates in the survival data analysis. For example, Morita, Lee and Mowday (1993) define a set of time-dependent dummy variables that allows assessment of the effects of changes in some predictor variables on dependent variable. For modeling the organizational characteristics - entry timing linkage, future researchers can collect further data indicating changes in organizational characteristics over time.

Managerial Implications

This research stresses to managers the importance and usefulness of organizational characteristics analysis. It demonstrates how internal factors influence a firm's entry timing decisions. Major competitors' entry timing usually can cause strong impacts on market structures and reshape situations of competition (Swann & Gill, 1993). When there are only limited market signals which provide competitors' entry information, an analysis of organizational characteristics, as shown in this research, may offer useful intelligence.

By analyzing their own organizations, managers can gain valuable insights about their entry behavior. This research shows that organizational characteristics can be important determinants to accelerate/decelerate entry. Therefore, managers, when trying to adjust their product introductions to emerging markets, have to pay attention to their organizational structures. For instance, in order for a firm to become a pioneer, a feasible opportunity may present itself. The occurrence of such an opportunity depends on the firm's own strategic emphasis, skill, and structure. Further, the rationale behind significant findings in this research also provide valuable directions in which managers can modify their organizations. For example, the linkage between TMT homogeneity and early entry implies the importance of team integration for organizational innovation. Also, the relationship between lower performance and early entry may suggest that

organizations that are more committed to the status quo and more satisfied with the present performance tend to prolong their entry.

Conclusions

This dissertation makes substantive contributions to the field of strategic management research, in terms of results, theory, methodology, and practical management knowledge. This research has made a significant contribution to a better understanding of a firm's timing of entry into an emerging product market. This study has shown that, at least to some significant degree, the entry timing decisions are endogenous. The empirical evidence of this research indicates that the firms with different organizational characteristics, including TMT trait and diversity, boundary spanning, performance, capital structure, and firm size, tend to pursue different entry timing strategies. This finding confirms previous theoretical arguments that a firm's timing of entry into an emerging product market is 'accelerated' or 'decelerated' depending on the firm's organizational characteristics.

This research draws on existing organizational theories to test new concepts. For modeling such complex firm behavior as entry timing decision, this study hypothesized the effects of organizational characteristics on entry timing based on a set of organizational theories, including agency theory, control theory, strategic choice, transaction cost economics. Kaplan (1964) argues that a hypothesis may be as much confirmed by fitting it into other theories as by fitting it to the facts, because the newly developed theory/model then enjoys the

conceptual coherence and support provided by evidence for all other theories.

This study makes a significant contribution to the research design for the endogenous entry timing research. To avoid the possible external validity problem that single-industry study may face, this research employs a multi-industry approach. Ten product categories selected for analysis in this study, based on four-digit Standard Industrial Classification (SIC), belong to nine industries. Furthermore, this research avoids the possible self-reported data bias that might qualify the conclusions of previous entry-timing/entry-order studies using cross-sectional databases (e.g., PIMS). Data for entry timing were collected from the time a product category was emerging. The entry events for each of the product categories were recorded through a process of assembling, critically examining, and summarizing the information of the past. The records of the past used in this study are all publicly available, published sources of information.

Another contribution of this research is that it has important implication for practitioners. This study gives top management new insight for examining their own organizational characteristics when trying to adjust their entry strategy, such as speeding product introduction to an emerging market. It also provides valuable directions in which managers can analyze competitors's entry behaviors through studying their organizational characteristics.

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